FH-OÖ Hagenberg/HSD
SDP3, WS 2026
Übung 3

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Beispiel 1 (24 Punkte) Player-Schnittstelle: Sie verwenden in Ihrer Firma HSDSoft einen MusicPlayer von der Firma MonkeySoft. Die öffentliche Schnittstelle des MusicPlayers sieht folgendermaßen aus und kann nicht verändert werden:

```
1     //starts playing with the actual song in list
2     void Start();
3     //stops playing
4     void Stop();
5     //switches to next song and starts at the end with first song
6     void SwitchNext();
7     //gets index of actual song
8     size_t GetActIndex() const;
9     //finds a song by name in playlist
10    bool Find(std::string const& name);
11    //gets count of songs in playlist
12    size_t GetCount() const;
13    //increases the volume relative to the actual volume
14    void IncreaseVol(size_t const vol);
15    //decreases the volume relative to the actual volume
16    void DecreaseVol(size_t const vol);
17    //adds a song to playlist
18    void Add(std::string const& name, size_t const dur);
```

Der MusicPlayer verwaltet Lieder und speichert den Namen und die Dauer jedes Liedes in Sekunden. Er kann gestartet und gestoppt werden und erlaubt das Verändern der Lautstärke. Die Lautstärke ist begrenzt mit 0 und maximal 100. Der Defaultwert für die Lautstärke liegt bei 15.

Zur Simulation liefert der Player je nach Aktion folgende Ausgaben auf der Konsole:

```
playing song number 1: Hells Bells (256 sec)
...
playing song number 4: Hawaguck (129 sec)
...
volume is now -> 70
song: Pulp Fiction not found!
stop song: Hells Bells (256 sec)
...
no song in playlist!
```

In weiterer Folge kaufen Sie einen VideoPlayer der Firma DonkeySoft mit folgender vorgegebenen Schnittstelle:

```
    //starts playing with the actual video in list
    void Play() const;
    //stops playing
    void Stop() const;
    //switches to first video in playlist and returns true, otherwise false if list is empty.
    bool First();
    //switches to next video in playlist and returns true, otherwise false if last song is reached.
    bool Next();
    //returns index of actual video
    size_t ActIndex() const;
    //returns name of actual video
    std::string const ActVideo() const;
    //sets volume (min volume=0 and max volume=50)
    void SetVolume(size_t const vol);
    //gets actual volume
    size_t GetVolume() const;
    //adds a video to playlist
    void Add(std::string const& name, size_t const dur, VideoFormat const& format);
}
```

Der VideoPlayer kann die Formate WMV, AVI und MKV abspielen. Er verwaltet Videos und speichert den Namen und die Dauer in Minuten. Er kann gestartet und gestoppt werden und erlaubt das Verändern der Lautstärke. Die Lautstärke ist begrenzt mit 0 und maximal 50. Der Defaultwert für die Lautstärke liegt bei 8.

Zur Simulation liefert der Player je nach Aktion folgende Ausgaben auf der Konsole:

```
playing video number 1: Die Sendung mit der Maus [duration -> 55 min], AVI-Format
...
playing video number 3: Freitag der 13te [duration -> 95 min], WMV-Format
...
volume is now -> 30
video: Hells Bells not found!
stop video: Pulp Fiction [duration -> 126 min], MKV-Format
...
no video in playlist!
```

Für einen Klienten soll nun nach außen folgende, unabhängige Schnittstelle zur Verfügung gestellt werden:

```
virtual void Play() = 0;
virtual void VolInc() = 0;
virtual void VolDec() = 0;
virtual void Stop() = 0;
virtual void Next() = 0;
virtual void Prev() = 0;
virtual void Select(std::string const& name) = 0;
```

Mit dieser Schnittstelle kann der Klient sowohl den MusicPlayer als auch den VideoPlayer verwenden. Die Methoden `VolInc()` und `VolDec()` erhöhen bzw. erniedrigen die Lautstärke um den Wert 1. `Next()` und `Prev()` schalten vor und zurück. `Select(...)` wählt ein Lied oder ein Video aus der Playliste aus und spielt es ab.

Implementieren Sie alle notwendigen Klassen (auch die Music/VideoPlayer-Klassen) und testen Sie diese entsprechend!

Implementieren Sie eine eigene `PlayerClient`-Klasse, die beide Player testet!

Achten Sie beim Design auf die Einhaltung der Design-Prinzipien und verwenden Sie ein entsprechendes Design-Pattern!

Allgemeine Hinweise: Legen Sie bei der Erstellung Ihrer Übung großen Wert auf eine **saubere Strukturierung** und auf eine **sorgfältige Ausarbeitung!** Dokumentieren Sie alle Schnittstellen und versehen Sie Ihre Algorithmen an entscheidenden Stellen ausführlich mit Kommentaren! Testen Sie ihre Implementierungen ausführlich! Geben Sie den **Testoutput** mit ab!



Systemdokumentation
Projekt Player-Schnittstelle

Version 1.0

T. Lechner, P. Ackermann

Hagenberg, 4. November 2026

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1 Organisatorisches

1.1 Team

- T. Lechner, Matr.-Nr.: S2510306017, E-Mail: s2510306017@students.fh-hagenberg.at
- P. Ackermann, Matr.-Nr.: S2510306001, E-Mail: s2510306001@students.fh-hagenberg.at

1.2 Aufteilung der Verantwortlichkeitsbereiche

- T. Lechner
  - Anforderungsdefinition
  - Implementierung und Unit-Test:
    - Video
    - Object
    - IMediaPlayer
    - VideoFormat
    - VideoPlayer
    - VideoPlayerAdapter
- P. Ackermann
  - Designentscheidungen
  - Implementierung und Unit-Test:
    - Song
    - MusicPlayer

* MusicPlayerAdapter
* PlayerClient

1.3 Aufwand

- T. Lechner: geschätzt 6 Ph / tatsächlich 7 Ph
- P. Ackermann: geschätzt 6 Ph / tatsächlich 6 Ph

2 Anforderungsdefinition (Systemspezifikation)

2.1 MonkeySoft

Es soll ein Musik Player implementiert werden. Dieser kann mehrere Songs, bestehend aus Titel und Dauer in Sekunden in seiner Playlist speichern.
Die Songs können abgespielt und das Abspielen wieder gestoppt werden. Es gibt auch die Möglichkeit, auf den nächsten Song zu wechseln, wenn alle Songs durch sind, wird wieder von vorne angefangen.
Der Musik Player verfügt über eine Lautstärke (0 bis 100, standardmäßig 15), die erhöht und verringert werden kann.
Neue Songs können hinzugefügt werden und es kann nach einem bestimmten Titel gesucht werden.
Die Anzahl der Songs in der Playlist und der Index des aktuell ausgewählten Songs kann ermittelt werden.

2.2 DonkeySoft

Es soll ein Video Player implementiert werden. Dieser kann mehrere Videos, bestehend aus Titel, Video-Format (AVI, MKV oder WMV) und Dauer in Minuten in seiner Playlist speichern.

Die Videos können abgespielt und das Abspielen wieder gestoppt werden. Es gibt auch die Möglichkeit, auf das nächste Video zu wechseln und wieder auf den Anfang zurückzuspringen.
Der Musik Player verfügt über eine Lautstärke (0 bis 50, standardmäßig 8), die erhöht und verringert werden kann.
Neue Videos können hinzugefügt werden.
Der Titel und Index des aktuell ausgewählten Videos kann auch ermittelt werden.

2.3 HSDSoft

Es soll für einen Klienten die oben genannten Player in einem einheitlichen Format mit einer gemeinsamen Schnittstelle bereitgestellt werden. Mit dieser Schnittstelle können beide Player gestartet und gestoppt werden. Es kann zwischen Liedern/Videos vor und zurück gewechselt werden. Wird über den Anfang hinaus zurückgewechselt, springt der Player wieder zum Ende. Wird das Ende überschritten, so fängt er wieder von vorne an. Die Lautstärke kann auch immer um 1 erhöht/verringert werden, bis die minimale/maximale Lautstärke erreicht ist. Es kann auch ein ganz bestimmter Titel (sofern dieser existiert) basierend auf dem Namen ausgewählt werden.

Das Hinzufügen neuer Medien ist über die Schnittstelle nicht möglich, sondern nur das Bedienen bereits ausgestatteter Player.

3 Systementwurf

3.1 Klassendiagramm

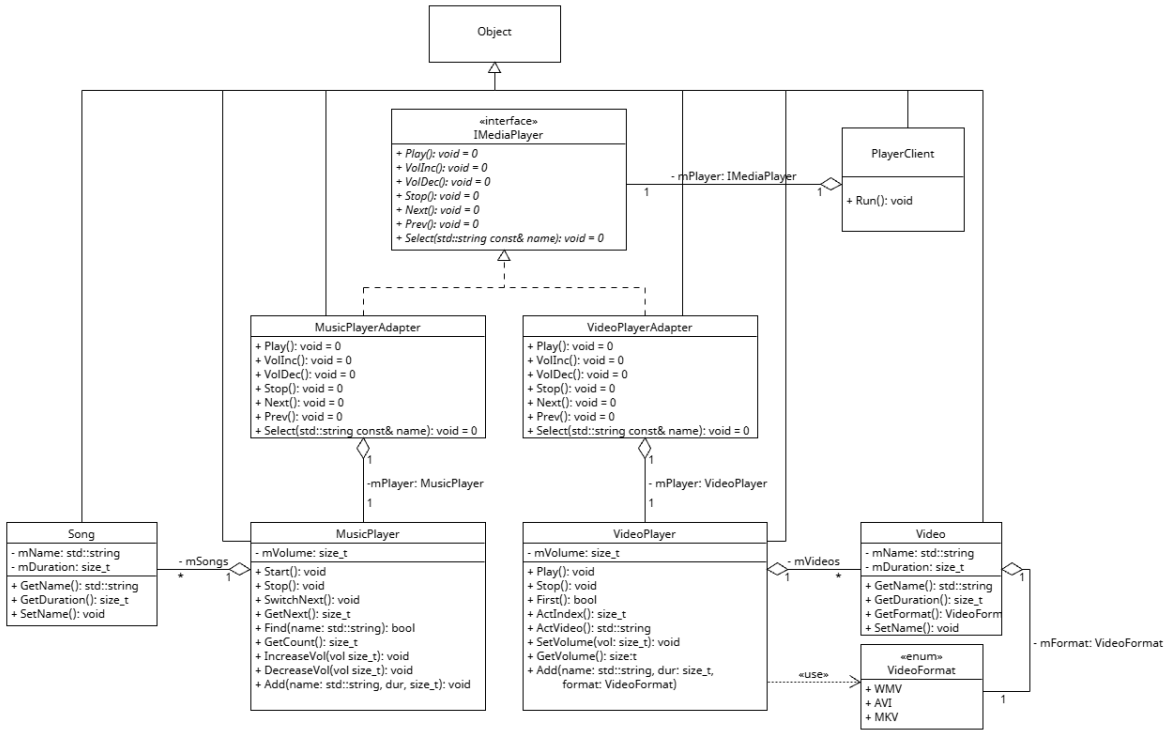


Abbildung 1: Klassendiagramm Player-Schnittstelle

3.2 Designentscheidungen

Um bei weiterer Ableitung der verwendeten Klassen bereits einen virtuellen Destruktor bereitzustellen, werden alle Klassen mit Ausnahme von Interfaces von einer gemeinsamen Object-Klasse mit virtuellem Destruktor abgeleitet.

Da die Klasse Client eine einheitliche Schnittstelle von sowohl dem Music-

Player als auch dem MediaPlayer benötigt, wird hier das Softwarepattern des Adapters angewandt. Hierbei werden für beide Klassen jeweils eine Adapter Klasse bereitgestellt, welche die gemeinsamen Funktionalitäten dem Client bereitstellen sollen. Diese Adapter erben von einem gemeinsamen Interface, damit auch später weitere Adapter für andere Klassen mit ähnlichen Funktionalitäten hinzugefügt werden können. Die Adapter Klassen müssen jeweils Objekte ihrer Klassen, die sie adaptieren aggregieren. So können auf deren Member über ihre Getter zugegriffen werden und auch andere bereitgestellte Methoden verwendet werden.

MusicPlayer:
Um den Zustand der Lautstärke, der über die vorgegebenen Methoden geändert werden soll zu speichern, wird ein Member verwendet. Ebenso wird ein Vector aus „Songs“ zur praktikablen Speicherung der Lieder verwendet. Die Klasse Songs wiederum beinhaltet Member zur Speicherung von Titel und Dauer. Es werden Getter-Methoden zum Lesen zu Verfügung gestellt. Das nachträgliche Ändern soll nur für den Namen möglich sein, da die Länge eines Songs als konstant angenommen wird.

VideoPlayer:
Ebenso wie im MusicPlayer muss auch hier der Zustand der Lautstärke und die einzelnen Videos gespeichert werden können. Für ein Video werden Member für Namen und Länge, aber auch für das Dateiformat benötigt. Die Werte für dieses Dateiformat, werden über ein Enum „Videoformat“ definiert.

Hinweis: Die Adapter für den MusicPlayer und den VideoPlayer werden wie in der Angabe gefordert über die Klasse PlayerClient ausführlich getestet. Da einige Methoden der Klassen, die die Adapter aggregieren von den Adaptermethoden nicht verwendet werden, müssen diese Klassen (MusicPlayer, Song, VideoPlayer, Video) eigenen Unit-Tests unterzogen werden.

4 Dokumentation der Komponenten (Klassen)

Die HTML-Startdatei befindet sich im Verzeichnis ./../doxy/html/index.html

5 Testprotokollierung

Die Tests wurden mittels Google Test Suite direkt in Visual Studio durchgeführt.

Listing 1: Systemdoku/Tests

```
Visual Leak Detector read settings from: C:\Program Files (x86)\
    Visual Leak Detector\vld.ini
Visual Leak Detector Version 2.5.1 installed.
Running main() from D:\a\_work\1\s\googletest\googletest\src\
    gtest_main.cc
[==========] Running 19 tests from 8 test cases.
[----------] Global test environment set-up.
[----------] 1 test from VideoFormat
[ RUN      ] VideoFormat.Test
[       OK ] VideoFormat.Test (0 ms)
[----------] 1 test from VideoFormat (0 ms total)

[----------] 1 test from Video
[ RUN      ] Video.Test
[       OK ] Video.Test (1 ms)
[----------] 1 test from Video (1 ms total)

[----------] 3 tests from VideoPlayer
[ RUN      ] VideoPlayer.Empty
no video in playlist!
no video in playlist!
volume is now -> 0
volume is now -> 50
volume is now -> 50
volume is now -> 50
[       OK ] VideoPlayer.Empty (4 ms)
[ RUN      ] VideoPlayer.OneVideo
stop video: Test [duration -> 123 min], AVI-Format
playing video number 1: Test [duration -> 123 min], AVI-Format
playing video number 1: Test [duration -> 123 min], AVI-Format
stop video: Test [duration -> 123 min], AVI-Format
playing video number 1: Test [duration -> 123 min], AVI-Format
stop video: Test [duration -> 123 min], AVI-Format
[       OK ] VideoPlayer.OneVideo (14 ms)
[ RUN      ] VideoPlayer.MultipleVideos
stop video: Video 1 [duration -> 1 min], AVI-Format
playing video number 1: Video 1 [duration -> 1 min], AVI-Format
```

```
playing video number 2: Video 2 [duration -> 2 min], MKV-Format
stop video: Video 2 [duration -> 2 min], MKV-Format
playing video number 3: Video 3 [duration -> 3 min], WMV-Format
stop video: Video 3 [duration -> 3 min], WMV-Format
playing video number 3: Video 3 [duration -> 3 min], WMV-Format
stop video: Video 3 [duration -> 3 min], WMV-Format
stop video: Video 1 [duration -> 1 min], AVI-Format
playing video number 1: Video 1 [duration -> 1 min], AVI-Format
[       OK ] VideoPlayer.MultipleVideos (21 ms)
[----------] 3 tests from VideoPlayer (40 ms total)

[----------] 1 test from Song
[ RUN      ] Song.Test
[       OK ] Song.Test (0 ms)
[----------] 1 test from Song (1 ms total)

[----------] 3 tests from MusicPlayer
[ RUN      ] MusicPlayer.Empty
no song in playlist!
no song in playlist!
volume is now -> 20
volume is now -> 20
volume is now -> 20
volume is now -> 15
volume is now -> 0
volume is now -> 100
[       OK ] MusicPlayer.Empty (17 ms)
[ RUN      ] MusicPlayer.OneSong
playing song number 1: Hells Bells (256 sec)
stop song: Hells Bells (256 sec)
playing song number 1: Hells Bells (256 sec)
stop song: Hells Bells (256 sec)
[       OK ] MusicPlayer.OneSong (8 ms)
[ RUN      ] MusicPlayer.MultipleSongs
playing song number 1: Hells Bells (256 sec)
stop song: Hells Bells (256 sec)
playing song number 2: Stereomadness (91 sec)
stop song: Stereomadness (91 sec)
playing song number 3: Clubstep (92 sec)
stop song: Clubstep (92 sec)
[       OK ] MusicPlayer.MultipleSongs (13 ms)
[----------] 3 tests from MusicPlayer (39 ms total)

[----------] 4 tests from VideoPlayerAdapter
[ RUN      ] VideoPlayerAdapter.nullptr
[       OK ] VideoPlayerAdapter.nullptr (0 ms)
[ RUN      ] VideoPlayerAdapter.Empty
### Test Stop/Play ###
no video in playlist!
no video in playlist!
Increase volume 110 times.
volume is now -> 9
volume is now -> 10
volume is now -> 11
volume is now -> 12
```

```
volume is now -> 13
volume is now -> 14
volume is now -> 15
volume is now -> 16
volume is now -> 17
volume is now -> 18
volume is now -> 19
volume is now -> 20
volume is now -> 21
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volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
Decrease volume 110 times.
volume is now -> 49
volume is now -> 48
volume is now -> 47
```

```
volume is now -> 46
volume is now -> 45
volume is now -> 44
volume is now -> 43
volume is now -> 42
volume is now -> 41
volume is now -> 40
volume is now -> 39
volume is now -> 38
volume is now -> 37
volume is now -> 36
volume is now -> 35
volume is now -> 34
volume is now -> 33
volume is now -> 32
volume is now -> 31
volume is now -> 30
volume is now -> 29
volume is now -> 28
volume is now -> 27
volume is now -> 26
volume is now -> 25
volume is now -> 24
volume is now -> 23
volume is now -> 22
volume is now -> 21
volume is now -> 20
volume is now -> 19
volume is now -> 18
volume is now -> 17
volume is now -> 16
volume is now -> 15
volume is now -> 14
volume is now -> 13
volume is now -> 12
volume is now -> 11
volume is now -> 10
volume is now -> 9
volume is now -> 8
volume is now -> 7
volume is now -> 6
volume is now -> 5
volume is now -> 4
volume is now -> 3
volume is now -> 2
volume is now -> 1
volume is now -> 0

### Test Next/Prev ###
- Wrap around to last:
no video in playlist!
- Wrap around to first:
no video in playlist!
- Switch through all entries:
no video in playlist!
```

```
no video in playlist!
no video in playlist!
no video in playlist!
no video in playlist!

### Test Select ###
-- Search for "Does not exist":
video: Does not exist not found!
no video in playlist!
no video in playlist!

-- Search for "Media 1":
video: Media 1 not found!
no video in playlist!
no video in playlist!

-- Search for "Media 2":
video: Media 2 not found!
no video in playlist!
no video in playlist!

-- Search for "Media 2" - Should find the same one again:
video: Media 2 not found!
no video in playlist!
no video in playlist!
[       OK ] VideoPlayerAdapter.Empty (13 ms)
[ RUN      ] VideoPlayerAdapter.One
### Test Stop/Play ###
stop video: Media 1 [duration -> 1 min], MKV-Format
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
Increase volume 110 times.
volume is now -> 9
volume is now -> 10
volume is now -> 11
volume is now -> 12
volume is now -> 13
volume is now -> 14
volume is now -> 15
volume is now -> 16
volume is now -> 17
volume is now -> 18
volume is now -> 19
volume is now -> 20
volume is now -> 21
volume is now -> 22
volume is now -> 23
volume is now -> 24
volume is now -> 25
volume is now -> 26
volume is now -> 27
volume is now -> 28
volume is now -> 29
volume is now -> 30
volume is now -> 31
volume is now -> 32
```

```
volume is now -> 33
volume is now -> 34
volume is now -> 35
volume is now -> 36
volume is now -> 37
volume is now -> 38
volume is now -> 39
volume is now -> 40
volume is now -> 41
volume is now -> 42
volume is now -> 43
volume is now -> 44
volume is now -> 45
volume is now -> 46
volume is now -> 47
volume is now -> 48
volume is now -> 49
volume is now -> 50
volume is now -> 50
volume is now -> 50
```

```
volume is now -> 50
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volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
Decrease volume 110 times.
volume is now -> 49
volume is now -> 48
volume is now -> 47
volume is now -> 46
volume is now -> 45
volume is now -> 44
volume is now -> 43
volume is now -> 42
volume is now -> 41
volume is now -> 40
volume is now -> 39
volume is now -> 38
volume is now -> 37
volume is now -> 36
volume is now -> 35
volume is now -> 34
volume is now -> 33
volume is now -> 32
volume is now -> 31
volume is now -> 30
volume is now -> 29
volume is now -> 28
volume is now -> 27
```

```
volume is now -> 26
volume is now -> 25
volume is now -> 24
volume is now -> 23
volume is now -> 22
volume is now -> 21
volume is now -> 20
volume is now -> 19
volume is now -> 18
volume is now -> 17
volume is now -> 16
volume is now -> 15
volume is now -> 14
volume is now -> 13
volume is now -> 12
volume is now -> 11
volume is now -> 10
volume is now -> 9
volume is now -> 8
volume is now -> 7
volume is now -> 6
volume is now -> 5
volume is now -> 4
volume is now -> 3
volume is now -> 2
volume is now -> 1
volume is now -> 0

### Test Next/Prev ###
- Wrap around to last:
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
- Wrap around to first:
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
- Switch through all entries:
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
playing video number 1: Media 1 [duration -> 1 min], MKV-Format

### Test Select ###
-- Search for "Does not exist":
video: Does not exist not found!
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
stop video: Media 1 [duration -> 1 min], MKV-Format

-- Search for "Media 1":
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
stop video: Media 1 [duration -> 1 min], MKV-Format

-- Search for "Media 2":
video: Media 2 not found!
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
stop video: Media 1 [duration -> 1 min], MKV-Format
```

```
-- Search for "Media 2" - Should find the same one again:
video: Media 2 not found!
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
stop video: Media 1 [duration -> 1 min], MKV-Format
[       OK ] VideoPlayerAdapter.One (7 ms)
[ RUN      ] VideoPlayerAdapter.Multiple
### Test Stop/Play ###
stop video: Media 1 [duration -> 1 min], MKV-Format
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
Increase volume 110 times.
volume is now -> 9
volume is now -> 10
volume is now -> 11
volume is now -> 12
volume is now -> 13
volume is now -> 14
volume is now -> 15
volume is now -> 16
volume is now -> 17
volume is now -> 18
volume is now -> 19
volume is now -> 20
volume is now -> 21
volume is now -> 22
volume is now -> 23
volume is now -> 24
volume is now -> 25
volume is now -> 26
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volume is now -> 34
volume is now -> 35
volume is now -> 36
volume is now -> 37
volume is now -> 38
volume is now -> 39
volume is now -> 40
volume is now -> 41
volume is now -> 42
volume is now -> 43
volume is now -> 44
volume is now -> 45
volume is now -> 46
volume is now -> 47
volume is now -> 48
volume is now -> 49
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
```

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volume is now -> 50
volume is now -> 50
volume is now -> 50
```

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volume is now -> 50
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volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
Decrease volume 110 times.
volume is now -> 49
volume is now -> 48
volume is now -> 47
volume is now -> 46
volume is now -> 45
volume is now -> 44
volume is now -> 43
volume is now -> 42
volume is now -> 41
volume is now -> 40
volume is now -> 39
volume is now -> 38
volume is now -> 37
volume is now -> 36
volume is now -> 35
volume is now -> 34
volume is now -> 33
volume is now -> 32
volume is now -> 31
volume is now -> 30
volume is now -> 29
volume is now -> 28
volume is now -> 27
volume is now -> 26
volume is now -> 25
volume is now -> 24
volume is now -> 23
volume is now -> 22
volume is now -> 21
volume is now -> 20
volume is now -> 19
volume is now -> 18
volume is now -> 17
volume is now -> 16
volume is now -> 15
volume is now -> 14
volume is now -> 13
volume is now -> 12
volume is now -> 11
volume is now -> 10
volume is now -> 9
volume is now -> 8
volume is now -> 7
volume is now -> 6
```

```
volume is now -> 5
volume is now -> 4
volume is now -> 3
volume is now -> 2
volume is now -> 1
volume is now -> 0

### Test Next/Prev ###
- Wrap around to last:
playing video number 4: Media 2 [duration -> 4 min], MKV-Format
- Wrap around to first:
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
- Switch through all entries:
playing video number 2: Media 2 [duration -> 2 min], AVI-Format
playing video number 3: Media 3 [duration -> 3 min], WMV-Format
playing video number 4: Media 2 [duration -> 4 min], MKV-Format
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
playing video number 2: Media 2 [duration -> 2 min], AVI-Format

### Test Select ###
-- Search for "Does not exist":
video: Does not exist not found!
playing video number 2: Media 2 [duration -> 2 min], AVI-Format
stop video: Media 2 [duration -> 2 min], AVI-Format

-- Search for "Media 1":
playing video number 1: Media 1 [duration -> 1 min], MKV-Format
stop video: Media 1 [duration -> 1 min], MKV-Format

-- Search for "Media 2":
playing video number 2: Media 2 [duration -> 2 min], AVI-Format
stop video: Media 2 [duration -> 2 min], AVI-Format

-- Search for "Media 2" - Should find the same one again:
playing video number 2: Media 2 [duration -> 2 min], AVI-Format
stop video: Media 2 [duration -> 2 min], AVI-Format
[       OK ] VideoPlayerAdapter.Multiple (6 ms)
[----------] 4 tests from VideoPlayerAdapter (27 ms total)

[----------] 4 tests from MusicPlayerAdapter
[ RUN      ] MusicPlayerAdapter.nullptr
[       OK ] MusicPlayerAdapter.nullptr (0 ms)
[ RUN      ] MusicPlayerAdapter.Empty
### Test Stop/Play ###
no song in playlist!
no song in playlist!
Increase volume 110 times.
volume is now -> 16
volume is now -> 17
volume is now -> 18
volume is now -> 19
volume is now -> 20
volume is now -> 21
volume is now -> 22
volume is now -> 23
```

```
volume is now -> 24
volume is now -> 25
volume is now -> 26
volume is now -> 27
volume is now -> 28
volume is now -> 29
volume is now -> 30
volume is now -> 31
volume is now -> 32
volume is now -> 33
volume is now -> 34
volume is now -> 35
volume is now -> 36
volume is now -> 37
volume is now -> 38
volume is now -> 39
volume is now -> 40
volume is now -> 41
volume is now -> 42
volume is now -> 43
volume is now -> 44
volume is now -> 45
volume is now -> 46
volume is now -> 47
volume is now -> 48
volume is now -> 49
volume is now -> 50
volume is now -> 51
volume is now -> 52
volume is now -> 53
volume is now -> 54
volume is now -> 55
volume is now -> 56
volume is now -> 57
volume is now -> 58
volume is now -> 59
volume is now -> 60
volume is now -> 61
volume is now -> 62
volume is now -> 63
volume is now -> 64
volume is now -> 65
volume is now -> 66
volume is now -> 67
volume is now -> 68
volume is now -> 69
volume is now -> 70
volume is now -> 71
volume is now -> 72
volume is now -> 73
volume is now -> 74
volume is now -> 75
volume is now -> 76
volume is now -> 77
volume is now -> 78
```

```
volume is now -> 79
volume is now -> 80
volume is now -> 81
volume is now -> 82
volume is now -> 83
volume is now -> 84
volume is now -> 85
volume is now -> 86
volume is now -> 87
volume is now -> 88
volume is now -> 89
volume is now -> 90
volume is now -> 91
volume is now -> 92
volume is now -> 93
volume is now -> 94
volume is now -> 95
volume is now -> 96
volume is now -> 97
volume is now -> 98
volume is now -> 99
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
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volume is now -> 100
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volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
Decrease volume 110 times.
volume is now -> 99
volume is now -> 98
volume is now -> 97
volume is now -> 96
volume is now -> 95
volume is now -> 94
volume is now -> 93
```

```
volume is now -> 92
volume is now -> 91
volume is now -> 90
volume is now -> 89
volume is now -> 88
volume is now -> 87
volume is now -> 86
volume is now -> 85
volume is now -> 84
volume is now -> 83
volume is now -> 82
volume is now -> 81
volume is now -> 80
volume is now -> 79
volume is now -> 78
volume is now -> 77
volume is now -> 76
volume is now -> 75
volume is now -> 74
volume is now -> 73
volume is now -> 72
volume is now -> 71
volume is now -> 70
volume is now -> 69
volume is now -> 68
volume is now -> 67
volume is now -> 66
volume is now -> 65
volume is now -> 64
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volume is now -> 62
volume is now -> 61
volume is now -> 60
volume is now -> 59
volume is now -> 58
volume is now -> 57
volume is now -> 56
volume is now -> 55
volume is now -> 54
volume is now -> 53
volume is now -> 52
volume is now -> 51
volume is now -> 50
volume is now -> 49
volume is now -> 48
volume is now -> 47
volume is now -> 46
volume is now -> 45
volume is now -> 44
volume is now -> 43
volume is now -> 42
volume is now -> 41
volume is now -> 40
volume is now -> 39
volume is now -> 38
```

```
volume is now -> 37
volume is now -> 36
volume is now -> 35
volume is now -> 34
volume is now -> 33
volume is now -> 32
volume is now -> 31
volume is now -> 30
volume is now -> 29
volume is now -> 28
volume is now -> 27
volume is now -> 26
volume is now -> 25
volume is now -> 24
volume is now -> 23
volume is now -> 22
volume is now -> 21
volume is now -> 20
volume is now -> 19
volume is now -> 18
volume is now -> 17
volume is now -> 16
volume is now -> 15
volume is now -> 14
volume is now -> 13
volume is now -> 12
volume is now -> 11
volume is now -> 10
volume is now -> 9
volume is now -> 8
volume is now -> 7
volume is now -> 6
volume is now -> 5
volume is now -> 4
volume is now -> 3
volume is now -> 2
volume is now -> 1
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0

### Test Next/Prev ###
- Wrap around to last:
no song in playlist!
- Wrap around to first:
no song in playlist!
- Switch through all entries:
```

```
no song in playlist!
no song in playlist!
no song in playlist!
no song in playlist!
no song in playlist!

### Test Select ###
-- Search for "Does not exist":
song: Does not exist not found!
no song in playlist!
no song in playlist!

-- Search for "Media 1":
song: Media 1 not found!
no song in playlist!
no song in playlist!

-- Search for "Media 2":
song: Media 2 not found!
no song in playlist!
no song in playlist!

-- Search for "Media 2" - Should find the same one again:
song: Media 2 not found!
no song in playlist!
no song in playlist!
[       OK ] MusicPlayerAdapter.Empty (6 ms)
[ RUN      ] MusicPlayerAdapter.One
### Test Stop/Play ###
stop song: Media 1 (1 sec)
playing song number 1: Media 1 (1 sec)
Increase volume 110 times.
volume is now -> 16
volume is now -> 17
volume is now -> 18
volume is now -> 19
volume is now -> 20
volume is now -> 21
volume is now -> 22
volume is now -> 23
volume is now -> 24
volume is now -> 25
volume is now -> 26
volume is now -> 27
volume is now -> 28
volume is now -> 29
volume is now -> 30
volume is now -> 31
volume is now -> 32
volume is now -> 33
volume is now -> 34
volume is now -> 35
volume is now -> 36
volume is now -> 37
volume is now -> 38
```

```
volume is now -> 39
volume is now -> 40
volume is now -> 41
volume is now -> 42
volume is now -> 43
volume is now -> 44
volume is now -> 45
volume is now -> 46
volume is now -> 47
volume is now -> 48
volume is now -> 49
volume is now -> 50
volume is now -> 51
volume is now -> 52
volume is now -> 53
volume is now -> 54
volume is now -> 55
volume is now -> 56
volume is now -> 57
volume is now -> 58
volume is now -> 59
volume is now -> 60
volume is now -> 61
volume is now -> 62
volume is now -> 63
volume is now -> 64
volume is now -> 65
volume is now -> 66
volume is now -> 67
volume is now -> 68
volume is now -> 69
volume is now -> 70
volume is now -> 71
volume is now -> 72
volume is now -> 73
volume is now -> 74
volume is now -> 75
volume is now -> 76
volume is now -> 77
volume is now -> 78
volume is now -> 79
volume is now -> 80
volume is now -> 81
volume is now -> 82
volume is now -> 83
volume is now -> 84
volume is now -> 85
volume is now -> 86
volume is now -> 87
volume is now -> 88
volume is now -> 89
volume is now -> 90
volume is now -> 91
volume is now -> 92
volume is now -> 93
```

```
volume is now -> 94
volume is now -> 95
volume is now -> 96
volume is now -> 97
volume is now -> 98
volume is now -> 99
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
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volume is now -> 100
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volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
Decrease volume 110 times.
volume is now -> 99
volume is now -> 98
volume is now -> 97
volume is now -> 96
volume is now -> 95
volume is now -> 94
volume is now -> 93
volume is now -> 92
volume is now -> 91
volume is now -> 90
volume is now -> 89
volume is now -> 88
volume is now -> 87
volume is now -> 86
volume is now -> 85
volume is now -> 84
volume is now -> 83
volume is now -> 82
volume is now -> 81
volume is now -> 80
volume is now -> 79
volume is now -> 78
```

```
volume is now -> 77
volume is now -> 76
volume is now -> 75
volume is now -> 74
volume is now -> 73
volume is now -> 72
volume is now -> 71
volume is now -> 70
volume is now -> 69
volume is now -> 68
volume is now -> 67
volume is now -> 66
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volume is now -> 64
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volume is now -> 62
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volume is now -> 53
volume is now -> 52
volume is now -> 51
volume is now -> 50
volume is now -> 49
volume is now -> 48
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volume is now -> 42
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volume is now -> 38
volume is now -> 37
volume is now -> 36
volume is now -> 35
volume is now -> 34
volume is now -> 33
volume is now -> 32
volume is now -> 31
volume is now -> 30
volume is now -> 29
volume is now -> 28
volume is now -> 27
volume is now -> 26
volume is now -> 25
volume is now -> 24
volume is now -> 23
```

```
volume is now -> 22
volume is now -> 21
volume is now -> 20
volume is now -> 19
volume is now -> 18
volume is now -> 17
volume is now -> 16
volume is now -> 15
volume is now -> 14
volume is now -> 13
volume is now -> 12
volume is now -> 11
volume is now -> 10
volume is now -> 9
volume is now -> 8
volume is now -> 7
volume is now -> 6
volume is now -> 5
volume is now -> 4
volume is now -> 3
volume is now -> 2
volume is now -> 1
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0

### Test Next/Prev ###
- Wrap around to last:
playing song number 1: Media 1 (1 sec)
- Wrap around to first:
playing song number 1: Media 1 (1 sec)
- Switch through all entries:
playing song number 1: Media 1 (1 sec)
playing song number 1: Media 1 (1 sec)
playing song number 1: Media 1 (1 sec)
playing song number 1: Media 1 (1 sec)
playing song number 1: Media 1 (1 sec)

### Test Select ###
-- Search for "Does not exist":
song: Does not exist not found!
playing song number 1: Media 1 (1 sec)
stop song: Media 1 (1 sec)

-- Search for "Media 1":
playing song number 1: Media 1 (1 sec)
stop song: Media 1 (1 sec)
```

```
-- Search for "Media 2":
song: Media 2 not found!
playing song number 1: Media 1 (1 sec)
stop song: Media 1 (1 sec)

-- Search for "Media 2" - Should find the same one again:
song: Media 2 not found!
playing song number 1: Media 1 (1 sec)
stop song: Media 1 (1 sec)
[       OK ] MusicPlayerAdapter.One (6 ms)
[ RUN      ] MusicPlayerAdapter.Multiple
### Test Stop/Play ###
stop song: Media 1 (1 sec)
playing song number 1: Media 1 (1 sec)
Increase volume 110 times.
volume is now -> 16
volume is now -> 17
volume is now -> 18
volume is now -> 19
volume is now -> 20
volume is now -> 21
volume is now -> 22
volume is now -> 23
volume is now -> 24
volume is now -> 25
volume is now -> 26
volume is now -> 27
volume is now -> 28
volume is now -> 29
volume is now -> 30
volume is now -> 31
volume is now -> 32
volume is now -> 33
volume is now -> 34
volume is now -> 35
volume is now -> 36
volume is now -> 37
volume is now -> 38
volume is now -> 39
volume is now -> 40
volume is now -> 41
volume is now -> 42
volume is now -> 43
volume is now -> 44
volume is now -> 45
volume is now -> 46
volume is now -> 47
volume is now -> 48
volume is now -> 49
volume is now -> 50
volume is now -> 51
volume is now -> 52
volume is now -> 53
volume is now -> 54
```

```
volume is now -> 55
volume is now -> 56
volume is now -> 57
volume is now -> 58
volume is now -> 59
volume is now -> 60
volume is now -> 61
volume is now -> 62
volume is now -> 63
volume is now -> 64
volume is now -> 65
volume is now -> 66
volume is now -> 67
volume is now -> 68
volume is now -> 69
volume is now -> 70
volume is now -> 71
volume is now -> 72
volume is now -> 73
volume is now -> 74
volume is now -> 75
volume is now -> 76
volume is now -> 77
volume is now -> 78
volume is now -> 79
volume is now -> 80
volume is now -> 81
volume is now -> 82
volume is now -> 83
volume is now -> 84
volume is now -> 85
volume is now -> 86
volume is now -> 87
volume is now -> 88
volume is now -> 89
volume is now -> 90
volume is now -> 91
volume is now -> 92
volume is now -> 93
volume is now -> 94
volume is now -> 95
volume is now -> 96
volume is now -> 97
volume is now -> 98
volume is now -> 99
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
```

```
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
Decrease volume 110 times.
volume is now -> 99
volume is now -> 98
volume is now -> 97
volume is now -> 96
volume is now -> 95
volume is now -> 94
volume is now -> 93
volume is now -> 92
volume is now -> 91
volume is now -> 90
volume is now -> 89
volume is now -> 88
volume is now -> 87
volume is now -> 86
volume is now -> 85
volume is now -> 84
volume is now -> 83
volume is now -> 82
volume is now -> 81
volume is now -> 80
volume is now -> 79
volume is now -> 78
volume is now -> 77
volume is now -> 76
volume is now -> 75
volume is now -> 74
volume is now -> 73
volume is now -> 72
volume is now -> 71
volume is now -> 70
volume is now -> 69
volume is now -> 68
volume is now -> 67
volume is now -> 66
volume is now -> 65
volume is now -> 64
volume is now -> 63
volume is now -> 62
```

```
volume is now -> 61
volume is now -> 60
volume is now -> 59
volume is now -> 58
volume is now -> 57
volume is now -> 56
volume is now -> 55
volume is now -> 54
volume is now -> 53
volume is now -> 52
volume is now -> 51
volume is now -> 50
volume is now -> 49
volume is now -> 48
volume is now -> 47
volume is now -> 46
volume is now -> 45
volume is now -> 44
volume is now -> 43
volume is now -> 42
volume is now -> 41
volume is now -> 40
volume is now -> 39
volume is now -> 38
volume is now -> 37
volume is now -> 36
volume is now -> 35
volume is now -> 34
volume is now -> 33
volume is now -> 32
volume is now -> 31
volume is now -> 30
volume is now -> 29
volume is now -> 28
volume is now -> 27
volume is now -> 26
volume is now -> 25
volume is now -> 24
volume is now -> 23
volume is now -> 22
volume is now -> 21
volume is now -> 20
volume is now -> 19
volume is now -> 18
volume is now -> 17
volume is now -> 16
volume is now -> 15
volume is now -> 14
volume is now -> 13
volume is now -> 12
volume is now -> 11
volume is now -> 10
volume is now -> 9
volume is now -> 8
volume is now -> 7
```

```
volume is now -> 6
volume is now -> 5
volume is now -> 4
volume is now -> 3
volume is now -> 2
volume is now -> 1
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0

### Test Next/Prev ###
- Wrap around to last:
playing song number 4: Media 2 (4 sec)
- Wrap around to first:
playing song number 1: Media 1 (1 sec)
- Switch through all entries:
playing song number 2: Media 2 (2 sec)
playing song number 3: Media 3 (3 sec)
playing song number 4: Media 2 (4 sec)
playing song number 1: Media 1 (1 sec)
playing song number 2: Media 2 (2 sec)

### Test Select ###
-- Search for "Does not exist":
song: Does not exist not found!
playing song number 2: Media 2 (2 sec)
stop song: Media 2 (2 sec)

-- Search for "Media 1":
playing song number 1: Media 1 (1 sec)
stop song: Media 1 (1 sec)

-- Search for "Media 2":
playing song number 2: Media 2 (2 sec)
stop song: Media 2 (2 sec)

-- Search for "Media 2" - Should find the same one again:
playing song number 2: Media 2 (2 sec)
stop song: Media 2 (2 sec)
[       OK ] MusicPlayerAdapter.Multiple (7 ms)
[----------] 4 tests from MusicPlayerAdapter (21 ms total)

[----------] 2 tests from PlayerClient
[ RUN      ] PlayerClient.nullptr
[       OK ] PlayerClient.nullptr (0 ms)
[ RUN      ] PlayerClient.SetPlayer
### Test Stop/Play ###
```

```
no song in playlist!
no song in playlist!
Increase volume 110 times.
volume is now -> 16
volume is now -> 17
volume is now -> 18
volume is now -> 19
volume is now -> 20
volume is now -> 21
volume is now -> 22
volume is now -> 23
volume is now -> 24
volume is now -> 25
volume is now -> 26
volume is now -> 27
volume is now -> 28
volume is now -> 29
volume is now -> 30
volume is now -> 31
volume is now -> 32
volume is now -> 33
volume is now -> 34
volume is now -> 35
volume is now -> 36
volume is now -> 37
volume is now -> 38
volume is now -> 39
volume is now -> 40
volume is now -> 41
volume is now -> 42
volume is now -> 43
volume is now -> 44
volume is now -> 45
volume is now -> 46
volume is now -> 47
volume is now -> 48
volume is now -> 49
volume is now -> 50
volume is now -> 51
volume is now -> 52
volume is now -> 53
volume is now -> 54
volume is now -> 55
volume is now -> 56
volume is now -> 57
volume is now -> 58
volume is now -> 59
volume is now -> 60
volume is now -> 61
volume is now -> 62
volume is now -> 63
volume is now -> 64
volume is now -> 65
volume is now -> 66
volume is now -> 67
```

```
volume is now -> 68
volume is now -> 69
volume is now -> 70
volume is now -> 71
volume is now -> 72
volume is now -> 73
volume is now -> 74
volume is now -> 75
volume is now -> 76
volume is now -> 77
volume is now -> 78
volume is now -> 79
volume is now -> 80
volume is now -> 81
volume is now -> 82
volume is now -> 83
volume is now -> 84
volume is now -> 85
volume is now -> 86
volume is now -> 87
volume is now -> 88
volume is now -> 89
volume is now -> 90
volume is now -> 91
volume is now -> 92
volume is now -> 93
volume is now -> 94
volume is now -> 95
volume is now -> 96
volume is now -> 97
volume is now -> 98
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volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
volume is now -> 100
```

```
volume is now -> 100
volume is now -> 100
volume is now -> 100
Decrease volume 110 times.
volume is now -> 99
volume is now -> 98
volume is now -> 97
volume is now -> 96
volume is now -> 95
volume is now -> 94
volume is now -> 93
volume is now -> 92
volume is now -> 91
volume is now -> 90
volume is now -> 89
volume is now -> 88
volume is now -> 87
volume is now -> 86
volume is now -> 85
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volume is now -> 83
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volume is now -> 80
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volume is now -> 77
volume is now -> 76
volume is now -> 75
volume is now -> 74
volume is now -> 73
volume is now -> 72
volume is now -> 71
volume is now -> 70
volume is now -> 69
volume is now -> 68
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volume is now -> 66
volume is now -> 65
volume is now -> 64
volume is now -> 63
volume is now -> 62
volume is now -> 61
volume is now -> 60
volume is now -> 59
volume is now -> 58
volume is now -> 57
volume is now -> 56
volume is now -> 55
volume is now -> 54
volume is now -> 53
volume is now -> 52
volume is now -> 51
volume is now -> 50
volume is now -> 49
```

```
volume is now -> 48
volume is now -> 47
volume is now -> 46
volume is now -> 45
volume is now -> 44
volume is now -> 43
volume is now -> 42
volume is now -> 41
volume is now -> 40
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volume is now -> 19
volume is now -> 18
volume is now -> 17
volume is now -> 16
volume is now -> 15
volume is now -> 14
volume is now -> 13
volume is now -> 12
volume is now -> 11
volume is now -> 10
volume is now -> 9
volume is now -> 8
volume is now -> 7
volume is now -> 6
volume is now -> 5
volume is now -> 4
volume is now -> 3
volume is now -> 2
volume is now -> 1
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0
```

```
volume is now -> 0
volume is now -> 0
volume is now -> 0
volume is now -> 0

### Test Next/Prev ###
- Wrap around to last:
no song in playlist!
- Wrap around to first:
no song in playlist!
- Switch through all entries:
no song in playlist!
no song in playlist!
no song in playlist!
no song in playlist!
no song in playlist!

### Test Select ###
-- Search for "Does not exist":
song: Does not exist not found!
no song in playlist!
no song in playlist!

-- Search for "Media 1":
song: Media 1 not found!
no song in playlist!
no song in playlist!

-- Search for "Media 2":
song: Media 2 not found!
no song in playlist!
no song in playlist!

-- Search for "Media 2" - Should find the same one again:
song: Media 2 not found!
no song in playlist!
no song in playlist!
### Test Stop/Play ###
no video in playlist!
no video in playlist!
Increase volume 110 times.
volume is now -> 9
volume is now -> 10
volume is now -> 11
volume is now -> 12
volume is now -> 13
volume is now -> 14
volume is now -> 15
volume is now -> 16
volume is now -> 17
volume is now -> 18
volume is now -> 19
volume is now -> 20
volume is now -> 21
volume is now -> 22
```

```
volume is now -> 23
volume is now -> 24
volume is now -> 25
volume is now -> 26
volume is now -> 27
volume is now -> 28
volume is now -> 29
volume is now -> 30
volume is now -> 31
volume is now -> 32
volume is now -> 33
volume is now -> 34
volume is now -> 35
volume is now -> 36
volume is now -> 37
volume is now -> 38
volume is now -> 39
volume is now -> 40
volume is now -> 41
volume is now -> 42
volume is now -> 43
volume is now -> 44
volume is now -> 45
volume is now -> 46
volume is now -> 47
volume is now -> 48
volume is now -> 49
volume is now -> 50
volume is now -> 50
volume is now -> 50
```

```
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
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volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
volume is now -> 50
Decrease volume 110 times.
volume is now -> 49
volume is now -> 48
volume is now -> 47
volume is now -> 46
volume is now -> 45
volume is now -> 44
volume is now -> 43
volume is now -> 42
volume is now -> 41
volume is now -> 40
volume is now -> 39
volume is now -> 38
volume is now -> 37
```

```
volume is now -> 36
volume is now -> 35
volume is now -> 34
volume is now -> 33
volume is now -> 32
volume is now -> 31
volume is now -> 30
volume is now -> 29
volume is now -> 28
volume is now -> 27
volume is now -> 26
volume is now -> 25
volume is now -> 24
volume is now -> 23
volume is now -> 22
volume is now -> 21
volume is now -> 20
volume is now -> 19
volume is now -> 18
volume is now -> 17
volume is now -> 16
volume is now -> 15
volume is now -> 14
volume is now -> 13
volume is now -> 12
volume is now -> 11
volume is now -> 10
volume is now -> 9
volume is now -> 8
volume is now -> 7
volume is now -> 6
volume is now -> 5
volume is now -> 4
volume is now -> 3
volume is now -> 2
volume is now -> 1
volume is now -> 0

### Test Next/Prev ###
- Wrap around to last:
no video in playlist!
- Wrap around to first:
no video in playlist!
- Switch through all entries:
no video in playlist!
no video in playlist!
no video in playlist!
no video in playlist!
no video in playlist!

### Test Select ###
-- Search for "Does not exist":
video: Does not exist not found!
no video in playlist!
no video in playlist!
```

```
-- Search for "Media 1":
video: Media 1 not found!
no video in playlist!
no video in playlist!

-- Search for "Media 2":
video: Media 2 not found!
no video in playlist!
no video in playlist!

-- Search for "Media 2" - Should find the same one again:
video: Media 2 not found!
no video in playlist!
no video in playlist!
[       OK ] PlayerClient.SetPlayer (10 ms)
[----------] 2 tests from PlayerClient (11 ms total)

[----------] Global test environment tear-down
[==========] 19 tests from 8 test cases ran. (147 ms total)
[  PASSED  ] 19 tests.
```

6 Quellcode

6.0.1 Object

Listing 2: "Beispiel01/Object.h"

```
1  #ifndef OBJECT_H
2  #define OBJECT_H
3
4  /// @author Thomas Lechner
5  /// @date 29.10.2026
6  /// @brief Base class of all base classes.
7  ///
8  /// Is the common root class of all classes. If there is no other
       base class,
9  /// this class is to be selected as the base class.
10 class Object
11 {
12 public:
13     /// @brief Virtual default destructor -> once virtual, always
           virtual.
14     ///
15     /// It is not necessary to declare another virtual destructor
16     /// in derived classes.
17     virtual ~Object() = default;
18
19 protected:
```

```
20      /// @brief Protected default constructor that makes the class
            abstract.
21      Object() = default;
22  };
23
24  #endif
```

6.1 MonkeySoft

6.1.1 Song

Listing 3: Beispiel01/Song.h

```
1   #ifndef SONG_H
2   #define SONG_H
3
4   #include "Object.h"
5
6   #include <string>
7
8
9   /// @author Philip Ackermann
10  /// @date 29.10.2026
11  /// @brief A song.
12  ///
13  /// Holds information about a song: name and duration in seconda.
14  class Song : public Object
15  {
16  public:
17      /// @brief CTor.
18      /// @param[in] name The name of the song.
19      /// @param[in] duration The duration of the song in seconds.
20      ///
21      /// Initializes the song class.
22      Song(std::string const& name, size_t const duration);
23
24      /// @brief Getter for name.
25      /// @returns The name of the song.
26      std::string GetName() const;
27
28      /// @brief Getter for duration.
29      /// @returns The duration of the song in seconds.
30      size_t GetDuration() const;
31
32      /// @brief Setter for name.
33      /// @param[in] name The new name of the song.
34      void SetName(std::string const& name);
35
36      /// @brief String representation of the song.
37      /// @returns The song as a formated string.
```

```
38      std::string ToString() const;
39  private:
40      /// @brief The name of the song.
41      std::string mName;
42      /// @brief The duration of the song in seconds.
43      size_t mDuration;
44  };
45
46  #endif
```

Listing 4: Beispiel01/Song.cpp

```
1   #include "Song.h"
2
3   Song::Song(std::string const& name, size_t const duration) :
4       mName{name},
5       mDuration{duration}{
6
7   }
8
9   std::string Song::GetName() const
10  {
11      return mName;
12  }
13
14  size_t Song::GetDuration() const
15  {
16      return mDuration;
17  }
18
19  void Song::SetName(std::string const& name)
20  {
21      mName = name;
22  }
23  std::string Song::ToString() const
24  {
25      return mName + " (" + std::to_string(mDuration) + " sec)";
26  }
```

6.1.2 MusicPlayer

Listing 5: Beispiel01/MusicPlayer.h

```
1   #ifndef MUSICPLAYER_H
2   #define MUSICPLAYER_H
3
4   #include "Object.h"
5   #include "Song.h"
6
7   #include <string>
```

```cpp
#include <vector>
#include <memory>

/// @author Philip Ackermann
/// @date 29.10.2026
/// @brief A music player for managing songs.
///
/// Holds multiple Song objects and lets you switch between them and play/stop them. Can adjust the volume of the song currently playing.
class MusicPlayer : public Object
{
public:
    /// @brief The default value for the volume.
    static constexpr size_t DEFAULT_VOLUME = 15;

    /// @brief The maximum value for the volume.
    static constexpr size_t MAX_VOLUME = 100;

    /// @brief std::unique_ptr of this class.
    using UPtr = std::unique_ptr<MusicPlayer>;

    /// @brief Starts playing the currently selected Song.
    void Start();

    /// @brief Stops playing the currently playing Song.
    void Stop();

    /// @brief Switches to the next Song in the playlist.
    void SwitchNext();

    /// @brief Get currently selected Song's index.
    /// @returns The index of the currently selected Song.
    size_t GetActIndex() const;

    /// @param[in] name Name of the song to look for.
    /// @returns If Song with name exists in playlist.
    bool Find(std::string const& name);

    /// @returns The number of songs in playlist.
    size_t GetCount() const;

    /// @brief Increases the volume.
    /// @param[in] vol The value to increase the volume.
    ///
    /// If the increased volume is higher than MAX_VOLUME, it is clamped to MAX_VOLUME.
    void IncreaseVol(size_t const vol);

    /// @brief Decreases the volume.
    /// @param[in] vol The value to decrease the volume.
    ///
    /// If the decreased volume is lower than 0, it is clamped to 0.
    void DecreaseVol(size_t const vol);
```

```
59
60      /// @brief Adds a Song to the playlist.
61      /// @param[in] name The name of the Song.
62      /// @param[in] dur The duration of the Song in seconds.
63      void Add(std::string const& name, size_t const dur);
64
65  private:
66      /// @brief The current volume.
67      size_t mVolume = MusicPlayer::DEFAULT_VOLUME;
68
69      /// @brief The playlist.
70      std::vector<Song> mSongs;
71
72      /// @brief The index of the currently selected Song.
73      size_t mActIndex = 0;
74  };
75
76  #endif
```

Listing 6: Beispiel01/MusicPlayer.cpp

```
1   #include "MusicPlayer.h"
2
3   #include <iostream>
4
5   static std::string const cNoSongs = "no song in playlist!";
6
7   void MusicPlayer::Start()
8   {
9       if (mSongs.empty()){
10          std::cout << cNoSongs << std::endl;
11      }
12      else
13      {
14          std::cout << "playing song number " << mActIndex + 1 << ":
                " << mSongs.at(mActIndex).ToString() << std::endl;
15      }
16  }
17  void MusicPlayer::Stop() {
18      if (mSongs.empty()) {
19          std::cout << cNoSongs << std::endl;
20      }
21      else
22      {
23          std::cout << "stop song: " << mSongs.at(mActIndex).
                ToString() << std::endl;
24      }
25  }
26
27  void MusicPlayer::SwitchNext()
28  {
29      size_t const numOfSongs = mSongs.size();
30      if (numOfSongs > 1)
31      {
32          // set index to first song if actIndex is at last one
```

```
33          mActIndex = (mActIndex + 1) % numOfSongs;
34      }
35
36  }
37
38  size_t MusicPlayer::GetActIndex() const
39  {
40      return mActIndex;
41  }
42
43  bool MusicPlayer::Find(std::string const& name)
44  {
45      for (size_t i = 0; i < mSongs.size(); i++) {
46          if (mSongs.at(i).GetName() == name)
47          {
48              mActIndex = i;
49              return true;
50          }
51      }
52      return false;
53  }
54
55  size_t MusicPlayer::GetCount() const
56  {
57      return mSongs.size();
58  }
59
60  void MusicPlayer::IncreaseVol(size_t const vol)
61  {
62      if (vol > MAX_VOLUME - mVolume) // prevent size_t overflow
63      {
64          mVolume = MAX_VOLUME;
65      }
66      else
67      {
68          mVolume = mVolume + vol;
69      }
70
71      std::cout << "volume is now -> " << mVolume << std::endl;
72  }
73
74  void MusicPlayer::DecreaseVol(size_t const vol)
75  {
76      if (vol > mVolume) // prevent size_t underflow
77      {
78          mVolume = 0;
79      }
80      else
81      {
82          mVolume = mVolume - vol;
83      }
84
85      std::cout << "volume is now -> " << mVolume << std::endl;
86  }
87
```

```
88 void MusicPlayer::Add(std::string const& name, size_t const dur)
89 {
90     mSongs.emplace_back(name, dur);
91 }
```

6.2 DonkeySoft

6.2.1 VideoFormat

Listing 7: Beispiel01/VideoFormat.h

```
1  #ifndef VIDEOFORMAT_H
2  #define VIDEOFORMAT_H
3
4  #include <string>
5
6  /// @author Thomas Lechner
7  /// @date 29.10.2026
8  /// @brief Video formats supported by VideoPlayer Video.
9  enum VideoFormat
10 {
11     WMV,
12     AVI,
13     MKV
14 };
15
16 /// @brief String representation of a VideoFormat.
17 /// @param[in] format The VideoFormat.
18 /// @returns The VideoFormat's string representation.
19 std::string to_string(VideoFormat const format);
20
21 #endif
```

Listing 8: Beispiel01/VideoFormat.cpp

```
1  #include "VideoFormat.h"
2
3  std::string to_string(VideoFormat const format)
4  {
5      switch (format)
6      {
7          case WMV:   return "WMV-Format";
8          case AVI:   return "AVI-Format";
9          case MKV:   return "MKV-Format";
10     }
11     return "Unknown-Format";
12 }
```

6.2.2 Video

Listing 9: Beispiel01/Video.h

```
1  #ifndef VIDEO_H
2  #define VIDEO_H
3
4  #include "Object.h"
5  #include "VideoFormat.h"
6
7  #include <string>
8
9
10 /// @author Thomas Lechner
11 /// @date 29.10.2026
12 /// @brief A video.
13 ///
14 /// Holds information about a video: name, duration in minutes and
       the VideoFormat.
15 class Video : public Object
16 {
17 public:
18     /// @brief CTor.
19     /// @param[in] name The name of the video.
20     /// @param[in] duration The duration of the video in minutes.
21     /// @param[in] format The video's VideoFormat.
22     ///
23     /// Initializes the Video class.
24     Video(std::string const& name, size_t const duration,
          VideoFormat const format);
25
26     /// @brief Getter for name.
27     /// @returns The name of the video.
28     std::string GetName() const;
29     /// @brief Getter for duration.
30     /// @returns The duration of the video in minutes.
31     size_t GetDuration() const;
32     /// @brief Getter for format.
33     /// @returns The VideoFormat of the video.
34     VideoFormat GetFormat() const;
35
36     /// @brief Setter for name.
37     /// @param[in] name The new name of the video.
38     void SetName(std::string const& name);
39
40     /// @brief String representation of the video.
41     /// @returns The video as a formated string.
42     std::string ToString() const;
43
44 private:
45     /// @brief The name of the video.
46     std::string mName;
47     /// @brief The duration of the video in minutes.
48     size_t mDuration;
49     /// @brief The VideoFormat of the video.
```

```
50      VideoFormat mFormat;
51  };
52
53  #endif
```

Listing 10: Beispiel01/Video.cpp

```
1   #include "Video.h"
2   #include "Song.h"
3
4   Video::Video(std::string const& name, size_t const duration,
        VideoFormat const format) :
5       mName{ name },
6       mDuration{ duration },
7       mFormat{ format }
8   {
9   }
10
11  std::string Video::GetName() const
12  {
13      return mName;
14  }
15
16  size_t Video::GetDuration() const
17  {
18      return mDuration;
19  }
20
21  VideoFormat Video::GetFormat() const
22  {
23      return mFormat;
24  }
25
26  void Video::SetName(std::string const& name)
27  {
28      mName = name;
29  }
30
31  std::string Video::ToString() const
32  {
33      return mName + " [duration -> " + std::to_string(mDuration) +
            " min], " + to_string(mFormat);
34  }
```

6.2.3 VideoPlayer

Listing 11: Beispiel01/VideoPlayer.h

```
1   #ifndef VIDEOPLAYER_H
2   #define VIDEOPLAYER_H
```

```
3
4  #include "Object.h"
5  #include "Video.h"
6
7  #include <string>
8  #include <vector>
9  #include <memory>
10
11
12 /// @author Thomas Lechner
13 /// @date 29.10.2026
14 /// @brief A video player for managing videos.
15 ///
16 /// Holds multiple Video objects and lets you switch between them
       and play/stop them. Can adjust the volume of the video
       currently playing.
17 class VideoPlayer : public Object
18 {
19 public:
20     /// @brief The default value for the volume.
21     static constexpr size_t DEFAULT_VOLUME = 8;
22     /// @brief The maximum value for the volume.
23     static constexpr size_t MAX_VOLUME = 50;
24
25     /// @brief std::unique_ptr of this class.
26     using UPtr = std::unique_ptr<VideoPlayer>;
27
28     /// @brief Starts playing the currently selected Video.
29     void Play() const;
30     /// @brief Stops playing the currently playing Video.
31     void Stop() const;
32
33     /// @brief Switches to the first Video in the playlist.
34     /// @returns false, when there are no Videos in the playlist;
           otherwise true.
35     bool First();
36     /// @brief Switches to the next Video in the playlist.
37     /// @returns false, when the last song is reached, otherwise
           true.
38     bool Next();
39
40     /// @brief Get currently selected Video's index.
41     /// @returns The index of the currently selected Video.
42     size_t ActIndex() const;
43     /// @brief Get currently selected Video's name.
44     /// @returns The name of the currently selected Video.
45     std::string const ActVideo() const;
46
47     /// @brief Sets the volume.
48     /// @param[in] vol The volume to change to.
49     ///
50     /// If the provided volume is bigger than MAX_VOLUME, it is
           clamped to MAX_VOLUME.
51     void SetVolume(size_t const vol);
52     /// @brief Gets the volume.
```

```
53      /// @returns The current volume.
54      size_t GetVolume() const;
55
56      /// @brief Adds a Video to the playlist.
57      /// @param[in] name The name of the video.
58      /// @param[in] dur The duration of the video in minutes.
59      /// @param[in] format The video's VideoFormat.
60      void Add(std::string const& name, size_t const dur,
            VideoFormat const& format);
61
62  private:
63      /// @brief The current volume.
64      size_t mVolume = VideoPlayer::DEFAULT_VOLUME;
65
66      /// @brief The playlist.
67      std::vector<Video> mVideos;
68
69      /// @brief The index of the currently selected Video.
70      size_t mCurIdx = 0;
71  };
72
73  #endif
```

Listing 12: Beispiel01/VideoPlayer.cpp

```
1   #include "VideoPlayer.h"
2
3   #include <iostream>
4
5   // Player output.
6   static std::string const cNoVideos = "no video in playlist!";
7
8   void VideoPlayer::Play() const
9   {
10      if (mVideos.empty())
11      {
12          std::cout << cNoVideos << std::endl;
13          return;
14      }
15
16      std::cout << "playing video number " << mCurIdx + 1 << ": " <<
            mVideos.at(mCurIdx).ToString() << std::endl;
17  }
18
19  void VideoPlayer::Stop() const
20  {
21      if (mVideos.empty())
22      {
23          std::cout << cNoVideos << std::endl;
24          return;
25      }
26
27      std::cout << "stop video: " << mVideos.at(mCurIdx).ToString()
            << std::endl;
28  }
```

```
29
30 bool VideoPlayer::First()
31 {
32     mCurIdx = 0;
33     return !mVideos.empty();
34 }
35
36 bool VideoPlayer::Next()
37 {
38     // No videos left?
39     if (mVideos.empty() || mCurIdx >= (mVideos.size() - 1)) return
           false;
40
41     ++mCurIdx;
42     return true;
43 }
44
45 size_t VideoPlayer::ActIndex() const
46 {
47     return mCurIdx;
48 }
49
50 std::string const VideoPlayer::ActVideo() const
51 {
52     if (mVideos.empty()) return "";
53
54     return mVideos.at(mCurIdx).GetName();
55 }
56
57 void VideoPlayer::SetVolume(size_t const vol)
58 {
59     mVolume = (vol > VideoPlayer::MAX_VOLUME) ? VideoPlayer::
          MAX_VOLUME : vol;
60
61     std::cout << "volume is now -> " << mVolume << std::endl;
62 }
63
64 size_t VideoPlayer::GetVolume() const
65 {
66     return mVolume;
67 }
68
69 void VideoPlayer::Add(std::string const& name, size_t const dur,
       VideoFormat const& format)
70 {
71     mVideos.emplace_back(name, dur, format);
72 }
```

6.3 HSDSoft

6.3.1 IMediaPlayer

Listing 13: "Beispiel01/IMediaPlayer.h"

```
1  #ifndef MEDIAPLAYER_H
2  #define MEDIAPLAYER_H
3
4  #include <string>
5  #include <memory>
6
7
8  /// @author Thomas Lechner
9  /// @date 29.10.2026
10 /// @brief Media player interface.
11 ///
12 /// Interface used to operate on a media player (volume control,
       play, stop, select).
13 class IMediaPlayer
14 {
15 public:
16     /// @brief std::unique_ptr of this class.
17     using UPtr = std::unique_ptr<IMediaPlayer>;
18
19     /// @brief Play the current media.
20     virtual void Play() = 0;
21     /// @brief Stop the current media.
22     virtual void Stop() = 0;
23
24     /// @brief Increase the volume by 1.
25     virtual void VolInc() = 0;
26     /// @brief Decrease the volume by 1.
27     virtual void VolDec() = 0;
28
29     /// @brief Select the next media.
30     virtual void Next() = 0;
31     /// @brief Select the previous media.
32     virtual void Prev() = 0;
33     /// @brief Select the media based on a name.
34     /// @param[in] name The name of the media to select.
35     virtual void Select(std::string const& name) = 0;
36
37     /// @brief Virtual DTor.
38     virtual ~IMediaPlayer() = default;
39 };
40
41 #endif
```

6.3.2 MusicPlayerAdapter

Listing 14: Beispiel01/MusicPlayerAdapter.h

```h
#ifndef MUSICPLAYERADAPTER_H
#define MUSICPLAYERADAPTER_H

#include "IMediaPlayer.h"
#include "MusicPlayer.h"

/// @author Philip Ackermann
/// @date 29.10.2026
/// @brief IMediaPlayer-Adapter for MusicPlayer.
///
/// Adapts a MusicPlayer to the IMediaPlayer interface.
class MusicPlayerAdapter : public Object, public IMediaPlayer
{
public:
    /// @brief CTor.
    /// @param[in] player The MusicPlayer to use.
    MusicPlayerAdapter(MusicPlayer::UPtr player);


    /// @brief Play the current Song.
    void Play() override;
    /// @brief Stop the current Song.
    void Stop() override;

    /// @brief Increase the volume by 1.
    void VolInc() override;
    /// @brief Decrease the volume by 1.
    void VolDec() override;

    /// @brief Select the next Song.
    ///
    /// Wraps around when the end is reached.
    void Next() override;
    /// @brief Select the previous Song.
    ///
    /// Wraps around when the beginning is reached.
    void Prev() override;
    /// @brief Select the Song based on a name.
    /// @param[in] name The name of the Song to select.
    void Select(std::string const& name) override;

private:
    /// @brief The MusicPlayer to use.
    MusicPlayer::UPtr mPlayer;
};

#endif
```

Listing 15: Beispiel01/MusicPlayerAdapter.cpp

```cpp
#include "MusicPlayerAdapter.h"
#include <iostream>

```

```
4  static std::string const cErrNullptr = "Provided player is nullptr
       ";
5
6  MusicPlayerAdapter::MusicPlayerAdapter(MusicPlayer::UPtr player)
7  {
8      if (player == nullptr) throw std::invalid_argument{
           cErrNullptr };
9
10     mPlayer = std::move(player);
11 }
12
13 void MusicPlayerAdapter::Play()
14 {
15     mPlayer->Start();
16 }
17
18 void MusicPlayerAdapter::Stop()
19 {
20     mPlayer->Stop();
21 }
22
23 void MusicPlayerAdapter::VolInc()
24 {
25     mPlayer->IncreaseVol(1);
26 }
27
28 void MusicPlayerAdapter::VolDec()
29 {
30     mPlayer->DecreaseVol(1);
31 }
32
33 void MusicPlayerAdapter::Next()
34 {
35     mPlayer->SwitchNext();
36 }
37
38 void MusicPlayerAdapter::Prev()
39 {
40     for (size_t i = mPlayer->GetCount(); i > 1; i--) {
41         //go forward size-1 times = go once backwards
42         mPlayer->SwitchNext();
43     }
44 }
45
46 void MusicPlayerAdapter::Select(std::string const& name)
47 {
48     if (!mPlayer->Find(name)) std::cout << "song: " << name << "
           not found!" << std::endl;
49 }
```

6.3.3 VideoPlayerAdapter

Listing 16: Beispiel01/VideoPlayerAdapter.h

```h
#ifndef VIDEOPLAYERADAPTER_H
#define VIDEOPLAYERADAPTER_H

#include "IMediaPlayer.h"
#include "VideoPlayer.h"

/// @author Thomas Lechner
/// @date 29.10.2026
/// @brief IMediaPlayer-Adapter for VideoPlayer.
///
/// Adapts a VideoPlayer to the IMediaPlayer interface.
class VideoPlayerAdapter : public Object, public IMediaPlayer
{
public:
    /// @brief CTor.
    /// @param[in] player The VideoPlayer to use.
    VideoPlayerAdapter(VideoPlayer::UPtr player);

    /// @brief Play the current Video.
    void Play() override;
    /// @brief Stop the current Video.
    void Stop() override;

    /// @brief Increase the volume by 1.
    void VolInc() override;
    /// @brief Decrease the volume by 1.
    void VolDec() override;

    /// @brief Select the next Video.
    ///
    /// Wraps around when the end is reached.
    void Next() override;
    /// @brief Select the previous Video.
    ///
    /// Does wrap around when the beginning is reached.
    void Prev() override;
    /// @brief Select the Video based on a name.
    /// @param[in] name The name of the Video to select.
    void Select(std::string const& name) override;

private:
    /// @brief The VideoPlayer to use.
    VideoPlayer::UPtr mPlayer;
};

#endif
```

Listing 17: Beispiel01/VideoPlayerAdapter.cpp

```cpp
#include "VideoPlayerAdapter.h"
```

```
2
3  #include <iostream>
4
5  static std::string const cErrNullptr = "Provided player is nullptr
       ";
6
7  VideoPlayerAdapter::VideoPlayerAdapter(VideoPlayer::UPtr player)
8  {
9      if (player == nullptr) throw std::invalid_argument{
           cErrNullptr };
10
11     mPlayer = std::move(player);
12 }
13
14 void VideoPlayerAdapter::Play()
15 {
16     mPlayer->Play();
17 }
18
19 void VideoPlayerAdapter::Stop()
20 {
21     mPlayer->Stop();
22 }
23
24 void VideoPlayerAdapter::VolInc()
25 {
26     mPlayer->SetVolume(mPlayer->GetVolume() + 1);
27 }
28
29 void VideoPlayerAdapter::VolDec()
30 {
31     size_t const vol = mPlayer->GetVolume(); // Avoid size_t
           underflow.
32     if (vol >= 1) mPlayer->SetVolume(vol - 1);
33 }
34
35 void VideoPlayerAdapter::Next()
36 {
37     // Wrap around when end is reached.
38     if (!mPlayer->Next()) mPlayer->First();
39 }
40
41 void VideoPlayerAdapter::Prev()
42 {
43     size_t const curIdx = mPlayer->ActIndex();
44     if (curIdx == 0) {
45         while (mPlayer->Next()) {}
46     }
47     else
48     {
49         mPlayer->First();
50         for (size_t i = 0; i < curIdx - 1; ++i)
51         {
52             mPlayer->Next();
53         }
```

```
54      }
55
56  }
57
58  void VideoPlayerAdapter::Select(std::string const& name)
59  {
60      size_t const oldIdx = mPlayer->ActIndex();
61
62      if (mPlayer->First()) // Player has videos.
63      {
64          do
65          {
66              if (mPlayer->ActVideo() == name) return; // Found!
67          } while (mPlayer->Next()); // Continue until end is
                reached.
68
69          // Was not found...
70          // ... go back to original idx.
71          mPlayer->First();
72          for (size_t i = 0; i < oldIdx; ++i)
73          {
74              mPlayer->Next();
75          }
76      }
77
78      std::cout << "video: " << name << " not found!" << std::endl;
79  }
```

6.3.4 PlayerClient

Listing 18: Beispiel01/PlayerClient.h

```
1   #ifndef PLAYERCLIENT_H
2   #define PLAYERCLIENT_H
3
4   #include "Object.h"
5   #include "IMediaPlayer.h"
6
7
8   /// @author Thomas Lechner
9   /// @date 03.11.2026
10  /// @brief A client of the IMediaPlayer interface.
11  ///
12  /// Tests multiple players by using their IMediaPlayer interface.
        Tests all IMediaPlayer methods.
13  class PlayerClient : public Object
14  {
15  public:
16      /// @brief CTor.
17      /// @param[in] player A UPtr to the IMediaPlayer to test.
18      PlayerClient(IMediaPlayer::UPtr player);
```

```
19
20      /// @brief Changes the IMediaPlayer the client operates on.
21      /// @param[in] player A UPtr to the IMediaPlayer to test.
22      void SetPlayer(IMediaPlayer::UPtr player);
23
24      /// @brief Run all tests for the selected IMediaPlayer.
25      void Run();
26
27  private:
28      /// @brief The IMediaPlayer under test.
29      IMediaPlayer::UPtr mPlayer;
30  };
31
32  #endif
```

Listing 19: Beispiel01/PlayerClient.cpp

```
1   #include "PlayerClient.h"
2
3   #include <stdexcept>
4   #include <iostream>
5
6   static std::string const cErrNullptr = "Provided player is nullptr
        ";
7
8   PlayerClient::PlayerClient(IMediaPlayer::UPtr player)
9   {
10      SetPlayer(std::move(player));
11  }
12
13  void PlayerClient::SetPlayer(IMediaPlayer::UPtr player)
14  {
15      if (player == nullptr) throw std::invalid_argument(cErrNullptr
            );
16
17      mPlayer = std::move(player);
18  }
19
20  void PlayerClient::Run()
21  {
22      if (mPlayer == nullptr) throw std::invalid_argument(
            cErrNullptr);
23
24      std::cout << "### Test Stop/Play ###\n";
25
26      mPlayer->Stop();
27      mPlayer->Play();
28
29      size_t const volChgAmount = 110;
30
31      std::cout << "Increase volume " << volChgAmount << " times.\n"
            ;
32      for (size_t i = 0; i < volChgAmount; ++i)
33      {
34          mPlayer->VolInc();
```

```
35      }
36
37      std::cout << "Decrease volume " << volChgAmount << " times.\n"
            ;
38      for (size_t i = 0; i < volChgAmount; ++i)
39      {
40          mPlayer->VolDec();
41      }
42
43      std::cout << "\n### Test Next/Prev ###\n";
44
45      std::cout << "- Wrap around to last:\n";
46      mPlayer->Prev();
47      mPlayer->Play();
48
49      std::cout << "- Wrap around to first:\n";
50      mPlayer->Next();
51      mPlayer->Play();
52
53      std::cout << "- Switch through all entries:\n";
54      for (size_t i = 0; i < 5; ++i)
55      {
56          mPlayer->Next();
57          mPlayer->Play();
58      }
59
60      std::cout << "\n### Test Select ###\n";
61
62      std::cout << "-- Search for \"Does not exist\":\n";
63      mPlayer->Select("Does not exist");
64      mPlayer->Play();
65      mPlayer->Stop();
66
67      std::cout << "\n-- Search for \"Media 1\":\n";
68      mPlayer->Select("Media 1");
69      mPlayer->Play();
70      mPlayer->Stop();
71
72      std::cout << "\n-- Search for \"Media 2\":\n";
73
74      mPlayer->Select("Media 2");
75      mPlayer->Play();
76      mPlayer->Stop();
77
78      std::cout << "\n-- Search for \"Media 2\" - Should find the
            same one again:\n";
79
80      mPlayer->Select("Media 2");
81      mPlayer->Play();
82      mPlayer->Stop();
83  }
```

6.4 Testtreiber

Listing 20: "Beispiel01/Test.cpp"

```
1  #include "gtest/gtest.h"
2
3  #include "Video.h"
4  #include "VideoPlayer.h"
5  #include "VideoPlayerAdapter.h"
6  #include "Song.h"
7  #include "MusicPlayer.h"
8  #include "MusicPlayerAdapter.h"
9  #include "PlayerClient.h"
10
11 #include <iostream>
12 #include <sstream>
13 #include <vld.h>
14
15 using namespace std;
16 using namespace testing::internal;
17
18 ////////////////////////////////////////////////
19
20 TEST(VideoFormat, Test) {
21     EXPECT_EQ(to_string(MKV), "MKV-Format");
22     EXPECT_EQ(to_string(AVI), "AVI-Format");
23     EXPECT_EQ(to_string(WMV), "WMV-Format");
24 }
25
26 ////////////////////////////////////////////////
27
28 TEST(Video, Test) {
29     Video vid1{ "Me at the zoo", 4, AVI };
30
31     EXPECT_EQ(vid1.GetName(), "Me at the zoo");
32     EXPECT_EQ(vid1.GetDuration(), 4);
33     EXPECT_EQ(vid1.GetFormat(), AVI);
34
35     EXPECT_EQ(vid1.ToString(), "Me at the zoo [duration -> 4 min],
           AVI-Format");
36
37     vid1.SetName("My new video");
38     EXPECT_EQ(vid1.GetName(), "My new video");
39
40     Video const vid2{ vid1 };
41     EXPECT_EQ(vid2.GetName(), vid1.GetName());
42     EXPECT_EQ(vid2.GetDuration(), vid1.GetDuration());
43     EXPECT_EQ(vid2.GetFormat(), vid1.GetFormat());
44     EXPECT_EQ(vid2.ToString(), vid1.ToString());
45
46     Video const vid3{ "YouTube video", 20, MKV };
47     vid1 = vid3;
48     EXPECT_EQ(vid3.GetName(), vid1.GetName());
49     EXPECT_EQ(vid3.GetDuration(), vid1.GetDuration());
50     EXPECT_EQ(vid3.GetFormat(), vid1.GetFormat());
```

```
51      EXPECT_EQ(vid3.ToString(), vid1.ToString());
52  }
53
54  ////////////////////////////////////////////////
55
56  TEST(VideoPlayer, Empty) {
57      VideoPlayer player;
58
59      std::string output = "";
60
61      CaptureStdout();
62      player.Play();
63      output = GetCapturedStdout();
64      cout << output;
65
66      EXPECT_EQ(output, "no video in playlist!\n");
67
68      CaptureStdout();
69      player.Stop();
70      output = GetCapturedStdout();
71      cout << output;
72
73      EXPECT_EQ(output, "no video in playlist!\n");
74
75      EXPECT_EQ(player.First(), false);
76      EXPECT_EQ(player.Next(), false);
77      EXPECT_EQ(player.ActIndex(), 0);
78      EXPECT_EQ(player.ActVideo(), "");
79
80      EXPECT_EQ(player.GetVolume(), VideoPlayer::DEFAULT_VOLUME);
81      player.SetVolume(0);
82      EXPECT_EQ(player.GetVolume(), 0);
83      player.SetVolume(50);
84      EXPECT_EQ(player.GetVolume(), 50);
85      player.SetVolume(VideoPlayer::MAX_VOLUME);
86      EXPECT_EQ(player.GetVolume(), VideoPlayer::MAX_VOLUME);
87      player.SetVolume(VideoPlayer::MAX_VOLUME + 1);
88      EXPECT_EQ(player.GetVolume(), VideoPlayer::MAX_VOLUME);
89  }
90
91  TEST(VideoPlayer, OneVideo) {
92      VideoPlayer player;
93      std::string output = "";
94
95      player.Add("Test", 123, VideoFormat::AVI);
96      EXPECT_EQ(player.ActIndex(), 0);
97      EXPECT_EQ(player.ActVideo(), "Test");
98
99      CaptureStdout();
100     player.Stop();
101     output = GetCapturedStdout();
102     cout << output;
103     EXPECT_EQ(output, "stop video: Test [duration -> 123 min], AVI
            -Format\n");
104
```

```
105     CaptureStdout();
106     player.Play();
107     output = GetCapturedStdout();
108     cout << output;
109     EXPECT_EQ(output, "playing video number 1: Test [duration ->
            123 min], AVI-Format\n");
110
111     player.Next();
112     EXPECT_EQ(player.ActIndex(), 0);
113     EXPECT_EQ(player.ActVideo(), "Test");
114
115     CaptureStdout();
116     player.Play();
117     output = GetCapturedStdout();
118     cout << output;
119     EXPECT_EQ(output, "playing video number 1: Test [duration ->
            123 min], AVI-Format\n");
120
121     CaptureStdout();
122     player.Stop();
123     output = GetCapturedStdout();
124     cout << output;
125     EXPECT_EQ(output, "stop video: Test [duration -> 123 min], AVI
            -Format\n");
126
127     player.First();
128     EXPECT_EQ(player.ActIndex(), 0);
129     EXPECT_EQ(player.ActVideo(), "Test");
130
131     CaptureStdout();
132     player.Play();
133     output = GetCapturedStdout();
134     cout << output;
135     EXPECT_EQ(output, "playing video number 1: Test [duration ->
            123 min], AVI-Format\n");
136
137     CaptureStdout();
138     player.Stop();
139     output = GetCapturedStdout();
140     cout << output;
141     EXPECT_EQ(output, "stop video: Test [duration -> 123 min], AVI
            -Format\n");
142 }
143
144 TEST(VideoPlayer, MultipleVideos) {
145     VideoPlayer player;
146     std::string output = "";
147
148     player.Add("Video 1", 1, VideoFormat::AVI);
149     player.Add("Video 2", 2, VideoFormat::MKV);
150     player.Add("Video 3", 3, VideoFormat::WMV);
151     EXPECT_EQ(player.ActIndex(), 0);
152     EXPECT_EQ(player.ActVideo(), "Video 1");
153
154     CaptureStdout();
```

```
155     player.Stop();
156     output = GetCapturedStdout();
157     cout << output;
158     EXPECT_EQ(output, "stop video: Video 1 [duration -> 1 min],
            AVI-Format\n");
159
160     CaptureStdout();
161     player.Play();
162     output = GetCapturedStdout();
163     cout << output;
164     EXPECT_EQ(output, "playing video number 1: Video 1 [duration
            -> 1 min], AVI-Format\n");
165
166     player.Next();
167     EXPECT_EQ(player.ActIndex(), 1);
168     EXPECT_EQ(player.ActVideo(), "Video 2");
169
170     CaptureStdout();
171     player.Play();
172     output = GetCapturedStdout();
173     cout << output;
174     EXPECT_EQ(output, "playing video number 2: Video 2 [duration
            -> 2 min], MKV-Format\n");
175
176     CaptureStdout();
177     player.Stop();
178     output = GetCapturedStdout();
179     cout << output;
180     EXPECT_EQ(output, "stop video: Video 2 [duration -> 2 min],
            MKV-Format\n");
181
182     player.Next();
183     EXPECT_EQ(player.ActIndex(), 2);
184     EXPECT_EQ(player.ActVideo(), "Video 3");
185
186     CaptureStdout();
187     player.Play();
188     output = GetCapturedStdout();
189     cout << output;
190     EXPECT_EQ(output, "playing video number 3: Video 3 [duration
            -> 3 min], WMV-Format\n");
191
192     CaptureStdout();
193     player.Stop();
194     output = GetCapturedStdout();
195     cout << output;
196     EXPECT_EQ(output, "stop video: Video 3 [duration -> 3 min],
            WMV-Format\n");
197
198     player.Next();
199     EXPECT_EQ(player.ActIndex(), 2);
200     EXPECT_EQ(player.ActVideo(), "Video 3");
201
202     CaptureStdout();
203     player.Play();
```

```
204     output = GetCapturedStdout();
205     cout << output;
206     EXPECT_EQ(output, "playing video number 3: Video 3 [duration
            -> 3 min], WMV-Format\n");
207
208     CaptureStdout();
209     player.Stop();
210     output = GetCapturedStdout();
211     cout << output;
212     EXPECT_EQ(output, "stop video: Video 3 [duration -> 3 min],
            WMV-Format\n");
213
214     player.First();
215     EXPECT_EQ(player.ActIndex(), 0);
216     EXPECT_EQ(player.ActVideo(), "Video 1");
217
218     CaptureStdout();
219     player.Stop();
220     output = GetCapturedStdout();
221     cout << output;
222     EXPECT_EQ(output, "stop video: Video 1 [duration -> 1 min],
            AVI-Format\n");
223
224     CaptureStdout();
225     player.Play();
226     output = GetCapturedStdout();
227     cout << output;
228     EXPECT_EQ(output, "playing video number 1: Video 1 [duration
            -> 1 min], AVI-Format\n");
229
230 }
231
232 ////////////////////////////////////////////////
233
234 TEST(Song, Test)
235 {
236     Song song1("Hells Bells", 256);
237
238     EXPECT_EQ(song1.GetName(), "Hells Bells");
239     EXPECT_EQ(song1.GetDuration(), 256);
240
241     EXPECT_EQ(song1.ToString(), "Hells Bells (256 sec)");
242
243     song1.SetName("Stereo Madness");
244     EXPECT_EQ(song1.GetName(), "Stereo Madness");
245
246     Song const song2{ song1 };
247     EXPECT_EQ(song1.GetDuration(), song2.GetDuration());
248     EXPECT_EQ(song1.GetName(), song2.GetName());
249     EXPECT_EQ(song1.ToString(), song2.ToString());
250
251     Song const song3("Deadlocked", 90);
252     song1 = song3;
253
254     EXPECT_EQ(song1.GetDuration(), song3.GetDuration());
```

```
255     EXPECT_EQ(song1.GetName(), song3.GetName());
256     EXPECT_EQ(song1.ToString(), song3.ToString());
257     EXPECT_EQ(song1.ToString(), song3.ToString());
258 }
259 
260 ////////////////////////////////////////////////
261 
262 TEST(MusicPlayer, Empty)
263 {
264     MusicPlayer player;
265 
266     string output = "";
267 
268     CaptureStdout();
269     player.Start();
270     output = GetCapturedStdout();
271     cout << output;
272 
273     EXPECT_EQ(output, "no song in playlist!\n");
274 
275     CaptureStdout();
276     player.Stop();
277     output = GetCapturedStdout();
278     cout << output;
279 
280     EXPECT_EQ(output, "no song in playlist!\n");
281 
282     CaptureStdout();
283     player.SwitchNext();
284     output = GetCapturedStdout();
285     cout << output;
286 
287     EXPECT_EQ(output, "");
288 
289     EXPECT_EQ(player.GetActIndex(), 0);
290 
291     EXPECT_EQ(player.Find("Hells Bells"), false);
292     EXPECT_EQ(player.GetActIndex(), 0);
293 
294     EXPECT_EQ(player.GetCount(), 0);
295 
296     CaptureStdout();
297     player.IncreaseVol(5);
298     output = GetCapturedStdout();
299     cout << output;
300     EXPECT_EQ(output, "volume is now -> " + to_string(MusicPlayer
        ::DEFAULT_VOLUME + 5) + '\n');
301 
302     CaptureStdout();
303     player.IncreaseVol(0);
304     output = GetCapturedStdout();
305     cout << output;
306     EXPECT_EQ(output, "volume is now -> " + to_string(MusicPlayer
        ::DEFAULT_VOLUME + 5) + '\n');
307 
```

```
308     CaptureStdout();
309     player.DecreaseVol(0);
310     output = GetCapturedStdout();
311     cout << output;
312     EXPECT_EQ(output, "volume is now -> " + to_string(MusicPlayer
            ::DEFAULT_VOLUME + 5) + '\n');
313
314     CaptureStdout();
315     player.DecreaseVol(5);
316     output = GetCapturedStdout();
317     cout << output;
318     EXPECT_EQ(output, "volume is now -> " + to_string(MusicPlayer
            ::DEFAULT_VOLUME) + '\n');
319
320     CaptureStdout();
321     player.DecreaseVol(50);
322     output = GetCapturedStdout();
323     cout << output;
324     EXPECT_EQ(output, "volume is now -> 0\n");
325
326     CaptureStdout();
327     player.IncreaseVol(MusicPlayer::MAX_VOLUME + 100);
328     output = GetCapturedStdout();
329     cout << output;
330     EXPECT_EQ(output, "volume is now -> " + to_string(MusicPlayer
            ::MAX_VOLUME) + '\n');
331
332 }
333
334 TEST(MusicPlayer, OneSong)
335 {
336     MusicPlayer player;
337
338     string output = "";
339
340     player.Add("Hells Bells", 256);
341     EXPECT_EQ(player.GetActIndex(), 0);
342
343     CaptureStdout();
344     player.Start();
345     output = GetCapturedStdout();
346     cout << output;
347     EXPECT_EQ(output, "playing song number 1: Hells Bells (256 sec
            )\n");
348
349     CaptureStdout();
350     player.Stop();
351     output = GetCapturedStdout();
352     cout << output;
353     EXPECT_EQ(output, "stop song: Hells Bells (256 sec)\n");
354
355
356     player.SwitchNext();
357     EXPECT_EQ(player.GetActIndex(), 0);
358
```

```
359
360     CaptureStdout();
361     player.Start();
362     output = GetCapturedStdout();
363     cout << output;
364     EXPECT_EQ(output, "playing song number 1: Hells Bells (256 sec
            )\n");
365
366     CaptureStdout();
367     player.Stop();
368     output = GetCapturedStdout();
369     cout << output;
370     EXPECT_EQ(output, "stop song: Hells Bells (256 sec)\n");
371
372
373     EXPECT_EQ(player.Find("Hells Bells"), true);
374     EXPECT_EQ(player.GetActIndex(), 0);
375
376     EXPECT_EQ(player.Find("StereoMadness"), false);
377     EXPECT_EQ(player.GetActIndex(), 0);
378 }
379
380 TEST(MusicPlayer, MultipleSongs)
381 {
382     MusicPlayer player;
383
384     string output = "";
385
386     player.Add("Hells Bells", 256);
387     player.Add("Stereomadness", 91);
388     player.Add("Clubstep", 92);
389     EXPECT_EQ(player.GetActIndex(), 0);
390
391     CaptureStdout();
392     player.Start();
393     output = GetCapturedStdout();
394     cout << output;
395     EXPECT_EQ(output, "playing song number 1: Hells Bells (256 sec
            )\n");
396
397     CaptureStdout();
398     player.Stop();
399     output = GetCapturedStdout();
400     cout << output;
401     EXPECT_EQ(output, "stop song: Hells Bells (256 sec)\n");
402
403
404     player.SwitchNext();
405     EXPECT_EQ(player.GetActIndex(), 1);
406
407
408     CaptureStdout();
409     player.Start();
410     output = GetCapturedStdout();
411     cout << output;
```

```
412     EXPECT_EQ(output, "playing song number 2: Stereomadness (91
            sec)\n");
413
414     CaptureStdout();
415     player.Stop();
416     output = GetCapturedStdout();
417     cout << output;
418     EXPECT_EQ(output, "stop song: Stereomadness (91 sec)\n");
419
420
421     player.SwitchNext();
422     EXPECT_EQ(player.GetActIndex(), 2);
423
424
425     CaptureStdout();
426     player.Start();
427     output = GetCapturedStdout();
428     cout << output;
429     EXPECT_EQ(output, "playing song number 3: Clubstep (92 sec)\n"
            );
430
431     CaptureStdout();
432     player.Stop();
433     output = GetCapturedStdout();
434     cout << output;
435     EXPECT_EQ(output, "stop song: Clubstep (92 sec)\n");
436
437     player.SwitchNext();
438     EXPECT_EQ(player.GetActIndex(), 0);
439
440
441     EXPECT_EQ(player.Find("Clubstep"), true);
442     EXPECT_EQ(player.GetActIndex(), 2);
443     EXPECT_EQ(player.Find("Slaughterhouse"), false);
444     EXPECT_EQ(player.GetActIndex(), 2);
445
446 }
447
448 ////////////////////////////////////////////////
449
450 TEST(VideoPlayerAdapter, nullptr) {
451     try {
452         VideoPlayerAdapter const player = VideoPlayerAdapter{
                nullptr };
453         EXPECT_EQ(1, 0); // Fail.
454     }
455     catch (std::invalid_argument const& ex) {
456         EXPECT_STREQ(ex.what(), "Provided player is nullptr");
457     }
458 }
459
460 TEST(VideoPlayerAdapter, Empty) {
461     std::string output = "";
462
463     CaptureStdout();
```

```
464     PlayerClient client{ std::make_unique<VideoPlayerAdapter>(std
            ::make_unique<VideoPlayer>()) };
465     client.Run();
466     output = GetCapturedStdout();
467     cout << output;
468
469     EXPECT_EQ(output, "### Test Stop/Play ###\nno video in
            playlist!\nno video in playlist!\nIncrease volume 110
            times.\nvolume is now -> 9\nvolume is now -> 10\nvolume is
             now -> 11\nvolume is now -> 12\nvolume is now -> 13\
            nvolume is now -> 14\nvolume is now -> 15\nvolume is now
            -> 16\nvolume is now -> 17\nvolume is now -> 18\nvolume is
             now -> 19\nvolume is now -> 20\nvolume is now -> 21\
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            nvolume is now -> 50\nDecrease volume 110 times.\nvolume
            is now -> 49\nvolume is now -> 48\nvolume is now -> 47\
            nvolume is now -> 46\nvolume is now -> 45\nvolume is now
            -> 44\nvolume is now -> 43\nvolume is now -> 42\nvolume is
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            nvolume is now -> 38\nvolume is now -> 37\nvolume is now
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        -> 20\nvolume is now -> 19\nvolume is now -> 18\nvolume is
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        nvolume is now -> 14\nvolume is now -> 13\nvolume is now
        -> 12\nvolume is now -> 11\nvolume is now -> 10\nvolume is
         now -> 9\nvolume is now -> 8\nvolume is now -> 7\nvolume
        is now -> 6\nvolume is now -> 5\nvolume is now -> 4\
        nvolume is now -> 3\nvolume is now -> 2\nvolume is now ->
        1\nvolume is now -> 0\n\n### Test Next/Prev ###\n- Wrap
        around to last:\nno video in playlist!\n- Wrap around to
        first:\nno video in playlist!\n- Switch through all
        entries:\nno video in playlist!\nno video in playlist!\nno
         video in playlist!\nno video in playlist!\nno video in
        playlist!\n\n### Test Select ###\n-- Search for \"Does not
         exist\":\nvideo: Does not exist not found!\nno video in
        playlist!\nno video in playlist!\n\n-- Search for \"Media
        1\":\nvideo: Media 1 not found!\nno video in playlist!\nno
         video in playlist!\n\n-- Search for \"Media 2\":\nvideo:
        Media 2 not found!\nno video in playlist!\nno video in
        playlist!\n\n-- Search for \"Media 2\" - Should find the
        same one again:\nvideo: Media 2 not found!\nno video in
        playlist!\nno video in playlist!\n");
}

TEST(VideoPlayerAdapter, One) {
    std::string output = "";

    VideoPlayer::UPtr player = std::make_unique<VideoPlayer>();
    player->Add("Media 1", 1, VideoFormat::MKV);

    CaptureStdout();
    PlayerClient client{ std::make_unique<VideoPlayerAdapter>(std
        ::move(player)) };
    client.Run();
    output = GetCapturedStdout();
    cout << output;

    EXPECT_EQ(output, "### Test Stop/Play ###\nstop video: Media 1
         [duration -> 1 min], MKV-Format\nplaying video number 1:
        Media 1 [duration -> 1 min], MKV-Format\nIncrease volume
        110 times.\nvolume is now -> 9\nvolume is now -> 10\
        nvolume is now -> 11\nvolume is now -> 12\nvolume is now
        -> 13\nvolume is now -> 14\nvolume is now -> 15\nvolume is
         now -> 16\nvolume is now -> 17\nvolume is now -> 18\
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nvolume is now -> 49\nvolume is now -> 48\nvolume is now
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 now -> 12\nvolume is now -> 11\nvolume is now -> 10\
nvolume is now -> 9\nvolume is now -> 8\nvolume is now ->
7\nvolume is now -> 6\nvolume is now -> 5\nvolume is now
-> 4\nvolume is now -> 3\nvolume is now -> 2\nvolume is
now -> 1\nvolume is now -> 0\n\n### Test Next/Prev ###\n-
Wrap around to last:\nplaying video number 1: Media 1 [
duration -> 1 min], MKV-Format\n- Wrap around to first:\
nplaying video number 1: Media 1 [duration -> 1 min], MKV-
Format\n- Switch through all entries:\nplaying video
```

```
        number 1: Media 1 [duration -> 1 min], MKV-Format\nplaying
         video number 1: Media 1 [duration -> 1 min], MKV-Format\
        nplaying video number 1: Media 1 [duration -> 1 min], MKV-
        Format\nplaying video number 1: Media 1 [duration -> 1 min
        ], MKV-Format\nplaying video number 1: Media 1 [duration
        -> 1 min], MKV-Format\n\n### Test Select ###\n-- Search
        for \"Does not exist\":\nvideo: Does not exist not found!\
        nplaying video number 1: Media 1 [duration -> 1 min], MKV-
        Format\nstop video: Media 1 [duration -> 1 min], MKV-
        Format\n\n-- Search for \"Media 1\":\nplaying video number
         1: Media 1 [duration -> 1 min], MKV-Format\nstop video:
        Media 1 [duration -> 1 min], MKV-Format\n\n-- Search for
        \"Media 2\":\nvideo: Media 2 not found!\nplaying video
        number 1: Media 1 [duration -> 1 min], MKV-Format\nstop
        video: Media 1 [duration -> 1 min], MKV-Format\n\n--
        Search for \"Media 2\" - Should find the same one again:\
        nvideo: Media 2 not found!\nplaying video number 1: Media
        1 [duration -> 1 min], MKV-Format\nstop video: Media 1 [
        duration -> 1 min], MKV-Format\n");
}

TEST(VideoPlayerAdapter, Multiple) {
    std::string output = "";

    VideoPlayer::UPtr player = std::make_unique<VideoPlayer>();
    player->Add("Media 1", 1, VideoFormat::MKV);
    player->Add("Media 2", 2, VideoFormat::AVI);
    player->Add("Media 3", 3, VideoFormat::WMV);
    player->Add("Media 2", 4, VideoFormat::MKV);

    CaptureStdout();
    PlayerClient client{ std::make_unique<VideoPlayerAdapter>(std
        ::move(player)) };
    client.Run();
    output = GetCapturedStdout();
    cout << output;

    EXPECT_EQ(output, "### Test Stop/Play ###\nstop video: Media 1
         [duration -> 1 min], MKV-Format\nplaying video number 1:
        Media 1 [duration -> 1 min], MKV-Format\nIncrease volume
        110 times.\nvolume is now -> 9\nvolume is now -> 10\
        nvolume is now -> 11\nvolume is now -> 12\nvolume is now
        -> 13\nvolume is now -> 14\nvolume is now -> 15\nvolume is
         now -> 16\nvolume is now -> 17\nvolume is now -> 18\
        nvolume is now -> 19\nvolume is now -> 20\nvolume is now
        -> 21\nvolume is now -> 22\nvolume is now -> 23\nvolume is
         now -> 24\nvolume is now -> 25\nvolume is now -> 26\
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        -> 29\nvolume is now -> 30\nvolume is now -> 31\nvolume is
         now -> 32\nvolume is now -> 33\nvolume is now -> 34\
        nvolume is now -> 35\nvolume is now -> 36\nvolume is now
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         now -> 40\nvolume is now -> 41\nvolume is now -> 42\
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nvolume is now -> 50\nvolume is now -> 50\nvolume is now
-> 50\nvolume is now -> 50\nDecrease volume 110 times.\
nvolume is now -> 49\nvolume is now -> 48\nvolume is now
-> 47\nvolume is now -> 46\nvolume is now -> 45\nvolume is
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-> 15\nvolume is now -> 14\nvolume is now -> 13\nvolume is
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nvolume is now -> 9\nvolume is now -> 8\nvolume is now ->
7\nvolume is now -> 6\nvolume is now -> 5\nvolume is now
-> 4\nvolume is now -> 3\nvolume is now -> 2\nvolume is
now -> 1\nvolume is now -> 0\n\n### Test Next/Prev ###\n-
Wrap around to last:\nplaying video number 4: Media 2 [
duration -> 4 min], MKV-Format\n- Wrap around to first:\
nplaying video number 1: Media 1 [duration -> 1 min], MKV-
Format\n- Switch through all entries:\nplaying video
number 2: Media 2 [duration -> 2 min], AVI-Format\nplaying
 video number 3: Media 3 [duration -> 3 min], WMV-Format\
nplaying video number 4: Media 2 [duration -> 4 min], MKV-
Format\nplaying video number 1: Media 1 [duration -> 1 min
], MKV-Format\nplaying video number 2: Media 2 [duration
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        -> 2 min], AVI-Format\n\n### Test Select ###\n-- Search
        for \"Does not exist\":\nvideo: Does not exist not found!\
        nplaying video number 2: Media 2 [duration -> 2 min], AVI-
        Format\nstop video: Media 2 [duration -> 2 min], AVI-
        Format\n\n-- Search for \"Media 1\":\nplaying video number
         1: Media 1 [duration -> 1 min], MKV-Format\nstop video:
        Media 1 [duration -> 1 min], MKV-Format\n\n-- Search for
        \"Media 2\":\nplaying video number 2: Media 2 [duration ->
         2 min], AVI-Format\nstop video: Media 2 [duration -> 2
        min], AVI-Format\n\n-- Search for \"Media 2\" - Should
        find the same one again:\nplaying video number 2: Media 2
        [duration -> 2 min], AVI-Format\nstop video: Media 2 [
        duration -> 2 min], AVI-Format\n");
}

////////////////////////////////////////////////

TEST(MusicPlayerAdapter, nullptr) {
    try {
        MusicPlayerAdapter const player = MusicPlayerAdapter{
            nullptr };
        EXPECT_EQ(1, 0); // Fail.
    }
    catch (std::invalid_argument const& ex) {
        EXPECT_STREQ(ex.what(), "Provided player is nullptr");
    }
}

TEST(MusicPlayerAdapter, Empty) {
    std::string output = "";

    CaptureStdout();
    PlayerClient client{ std::make_unique<MusicPlayerAdapter>(std
        ::make_unique<MusicPlayer>()) };
    client.Run();
    output = GetCapturedStdout();
    cout << output;

    EXPECT_EQ(output, "### Test Stop/Play ###\nno song in playlist
        !\nno song in playlist!\nIncrease volume 110 times.\
        nvolume is now -> 16\nvolume is now -> 17\nvolume is now
        -> 18\nvolume is now -> 19\nvolume is now -> 20\nvolume is
         now -> 21\nvolume is now -> 22\nvolume is now -> 23\
        nvolume is now -> 24\nvolume is now -> 25\nvolume is now
        -> 26\nvolume is now -> 27\nvolume is now -> 28\nvolume is
         now -> 29\nvolume is now -> 30\nvolume is now -> 31\
        nvolume is now -> 32\nvolume is now -> 33\nvolume is now
        -> 34\nvolume is now -> 35\nvolume is now -> 36\nvolume is
         now -> 37\nvolume is now -> 38\nvolume is now -> 39\
        nvolume is now -> 40\nvolume is now -> 41\nvolume is now
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         now -> 45\nvolume is now -> 46\nvolume is now -> 47\
        nvolume is now -> 48\nvolume is now -> 49\nvolume is now
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         now -> 53\nvolume is now -> 54\nvolume is now -> 55\
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nvolume is now -> 100\nvolume is now -> 100\nDecrease
volume 110 times.\nvolume is now -> 99\nvolume is now ->
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now -> 95\nvolume is now -> 94\nvolume is now -> 93\
nvolume is now -> 92\nvolume is now -> 91\nvolume is now
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        -> 26\nvolume is now -> 25\nvolume is now -> 24\nvolume is
         now -> 23\nvolume is now -> 22\nvolume is now -> 21\
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        -> 18\nvolume is now -> 17\nvolume is now -> 16\nvolume is
         now -> 15\nvolume is now -> 14\nvolume is now -> 13\
        nvolume is now -> 12\nvolume is now -> 11\nvolume is now
        -> 10\nvolume is now -> 9\nvolume is now -> 8\nvolume is
        now -> 7\nvolume is now -> 6\nvolume is now -> 5\nvolume
        is now -> 4\nvolume is now -> 3\nvolume is now -> 2\
        nvolume is now -> 1\nvolume is now -> 0\nvolume is now ->
        0\nvolume is now -> 0\nvolume is now -> 0\nvolume is now
        -> 0\nvolume is now -> 0\nvolume is now -> 0\nvolume is
        now -> 0\nvolume is now -> 0\nvolume is now -> 0\nvolume
        is now -> 0\n\n### Test Next/Prev ###\n- Wrap around to
        last:\nno song in playlist!\n- Wrap around to first:\nno
        song in playlist!\n- Switch through all entries:\nno song
        in playlist!\nno song in playlist!\nno song in playlist!\
        nno song in playlist!\nno song in playlist!\n\n### Test
        Select ###\n-- Search for \"Does not exist\":\nsong: Does
        not exist not found!\nno song in playlist!\nno song in
        playlist!\n\n-- Search for \"Media 1\":\nsong: Media 1 not
         found!\nno song in playlist!\nno song in playlist!\n\n--
        Search for \"Media 2\":\nsong: Media 2 not found!\nno song
         in playlist!\nno song in playlist!\n\n-- Search for \"
        Media 2\" - Should find the same one again:\nsong: Media 2
         not found!\nno song in playlist!\nno song in playlist!\n"
        );
}

TEST(MusicPlayerAdapter, One) {
    std::string output = "";

    MusicPlayer::UPtr player = std::make_unique<MusicPlayer>();
    player->Add("Media 1", 1);

    CaptureStdout();
    PlayerClient client{ std::make_unique<MusicPlayerAdapter>(std
        ::move(player)) };
    client.Run();
    output = GetCapturedStdout();
    cout << output;

    EXPECT_EQ(output, "### Test Stop/Play ###\nstop song: Media 1
        (1 sec)\nplaying song number 1: Media 1 (1 sec)\nIncrease
        volume 110 times.\nvolume is now -> 16\nvolume is now ->
        17\nvolume is now -> 18\nvolume is now -> 19\nvolume is
        now -> 20\nvolume is now -> 21\nvolume is now -> 22\
        nvolume is now -> 23\nvolume is now -> 24\nvolume is now
        -> 25\nvolume is now -> 26\nvolume is now -> 27\nvolume is
         now -> 28\nvolume is now -> 29\nvolume is now -> 30\
        nvolume is now -> 31\nvolume is now -> 32\nvolume is now
        -> 33\nvolume is now -> 34\nvolume is now -> 35\nvolume is
         now -> 36\nvolume is now -> 37\nvolume is now -> 38\
        nvolume is now -> 39\nvolume is now -> 40\nvolume is now
        -> 41\nvolume is now -> 42\nvolume is now -> 43\nvolume is
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 now -> 44\nvolume is now -> 45\nvolume is now -> 46\
nvolume is now -> 47\nvolume is now -> 48\nvolume is now
-> 49\nvolume is now -> 50\nvolume is now -> 51\nvolume is
 now -> 52\nvolume is now -> 53\nvolume is now -> 54\
nvolume is now -> 55\nvolume is now -> 56\nvolume is now
-> 57\nvolume is now -> 58\nvolume is now -> 59\nvolume is
 now -> 60\nvolume is now -> 61\nvolume is now -> 62\
nvolume is now -> 63\nvolume is now -> 64\nvolume is now
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-> 73\nvolume is now -> 74\nvolume is now -> 75\nvolume is
 now -> 76\nvolume is now -> 77\nvolume is now -> 78\
nvolume is now -> 79\nvolume is now -> 80\nvolume is now
-> 81\nvolume is now -> 82\nvolume is now -> 83\nvolume is
 now -> 84\nvolume is now -> 85\nvolume is now -> 86\
nvolume is now -> 87\nvolume is now -> 88\nvolume is now
-> 89\nvolume is now -> 90\nvolume is now -> 91\nvolume is
 now -> 92\nvolume is now -> 93\nvolume is now -> 94\
nvolume is now -> 95\nvolume is now -> 96\nvolume is now
-> 97\nvolume is now -> 98\nvolume is now -> 99\nvolume is
 now -> 100\nvolume is now -> 100\nvolume is now -> 100\
nvolume is now -> 100\nvolume is now -> 100\nvolume is now
 -> 100\nvolume is now -> 100\nvolume is now -> 100\
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nvolume is now -> 100\nvolume is now -> 100\nvolume is now
 -> 100\nvolume is now -> 100\nvolume is now -> 100\
nvolume is now -> 100\nvolume is now -> 100\nvolume is now
 -> 100\nvolume is now -> 100\nvolume is now -> 100\
nvolume is now -> 100\nvolume is now -> 100\nvolume is now
 -> 100\nDecrease volume 110 times.\nvolume is now -> 99\
nvolume is now -> 98\nvolume is now -> 97\nvolume is now
-> 96\nvolume is now -> 95\nvolume is now -> 94\nvolume is
 now -> 93\nvolume is now -> 92\nvolume is now -> 91\
nvolume is now -> 90\nvolume is now -> 89\nvolume is now
-> 88\nvolume is now -> 87\nvolume is now -> 86\nvolume is
 now -> 85\nvolume is now -> 84\nvolume is now -> 83\
nvolume is now -> 82\nvolume is now -> 81\nvolume is now
-> 80\nvolume is now -> 79\nvolume is now -> 78\nvolume is
 now -> 77\nvolume is now -> 76\nvolume is now -> 75\
nvolume is now -> 74\nvolume is now -> 73\nvolume is now
-> 72\nvolume is now -> 71\nvolume is now -> 70\nvolume is
 now -> 69\nvolume is now -> 68\nvolume is now -> 67\
nvolume is now -> 66\nvolume is now -> 65\nvolume is now
-> 64\nvolume is now -> 63\nvolume is now -> 62\nvolume is
 now -> 61\nvolume is now -> 60\nvolume is now -> 59\
nvolume is now -> 58\nvolume is now -> 57\nvolume is now
-> 56\nvolume is now -> 55\nvolume is now -> 54\nvolume is
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nvolume is now -> 50\nvolume is now -> 49\nvolume is now
-> 48\nvolume is now -> 47\nvolume is now -> 46\nvolume is
 now -> 45\nvolume is now -> 44\nvolume is now -> 43\
nvolume is now -> 42\nvolume is now -> 41\nvolume is now
-> 40\nvolume is now -> 39\nvolume is now -> 38\nvolume is
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            now -> 37\nvolume is now -> 36\nvolume is now -> 35\
           nvolume is now -> 34\nvolume is now -> 33\nvolume is now
           -> 32\nvolume is now -> 31\nvolume is now -> 30\nvolume is
            now -> 29\nvolume is now -> 28\nvolume is now -> 27\
           nvolume is now -> 26\nvolume is now -> 25\nvolume is now
           -> 24\nvolume is now -> 23\nvolume is now -> 22\nvolume is
            now -> 21\nvolume is now -> 20\nvolume is now -> 19\
           nvolume is now -> 18\nvolume is now -> 17\nvolume is now
           -> 16\nvolume is now -> 15\nvolume is now -> 14\nvolume is
            now -> 13\nvolume is now -> 12\nvolume is now -> 11\
           nvolume is now -> 10\nvolume is now -> 9\nvolume is now ->
            8\nvolume is now -> 7\nvolume is now -> 6\nvolume is now
           -> 5\nvolume is now -> 4\nvolume is now -> 3\nvolume is
           now -> 2\nvolume is now -> 1\nvolume is now -> 0\nvolume
           is now -> 0\nvolume is now -> 0\nvolume is now -> 0\
           nvolume is now -> 0\nvolume is now -> 0\nvolume is now ->
           0\nvolume is now -> 0\nvolume is now -> 0\nvolume is now
           -> 0\nvolume is now -> 0\n\n### Test Next/Prev ###\n- Wrap
            around to last:\nplaying song number 1: Media 1 (1 sec)\n
           - Wrap around to first:\nplaying song number 1: Media 1 (1
            sec)\n- Switch through all entries:\nplaying song number
           1: Media 1 (1 sec)\nplaying song number 1: Media 1 (1 sec)
           \nplaying song number 1: Media 1 (1 sec)\nplaying song
           number 1: Media 1 (1 sec)\nplaying song number 1: Media 1
           (1 sec)\n\n### Test Select ###\n-- Search for \"Does not
           exist\":\nsong: Does not exist not found!\nplaying song
           number 1: Media 1 (1 sec)\nstop song: Media 1 (1 sec)\n\n
           -- Search for \"Media 1\":\nplaying song number 1: Media 1
            (1 sec)\nstop song: Media 1 (1 sec)\n\n-- Search for \"
           Media 2\":\nsong: Media 2 not found!\nplaying song number
           1: Media 1 (1 sec)\nstop song: Media 1 (1 sec)\n\n--
           Search for \"Media 2\" - Should find the same one again:\
           nsong: Media 2 not found!\nplaying song number 1: Media 1
           (1 sec)\nstop song: Media 1 (1 sec)\n");
}

TEST(MusicPlayerAdapter, Multiple) {
    std::string output = "";

    MusicPlayer::UPtr player = std::make_unique<MusicPlayer>();
    player->Add("Media 1", 1);
    player->Add("Media 2", 2);
    player->Add("Media 3", 3);
    player->Add("Media 2", 4);

    CaptureStdout();
    PlayerClient client{ std::make_unique<MusicPlayerAdapter>(std
        ::move(player)) };
    client.Run();
    output = GetCapturedStdout();
    cout << output;

    EXPECT_EQ(output, "### Test Stop/Play ###\nstop song: Media 1
        (1 sec)\nplaying song number 1: Media 1 (1 sec)\nIncrease
        volume 110 times.\nvolume is now -> 16\nvolume is now ->
```

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17\nvolume is now -> 18\nvolume is now -> 19\nvolume is
now -> 20\nvolume is now -> 21\nvolume is now -> 22\
nvolume is now -> 23\nvolume is now -> 24\nvolume is now
-> 25\nvolume is now -> 26\nvolume is now -> 27\nvolume is
 now -> 28\nvolume is now -> 29\nvolume is now -> 30\
nvolume is now -> 31\nvolume is now -> 32\nvolume is now
-> 33\nvolume is now -> 34\nvolume is now -> 35\nvolume is
 now -> 36\nvolume is now -> 37\nvolume is now -> 38\
nvolume is now -> 39\nvolume is now -> 40\nvolume is now
-> 41\nvolume is now -> 42\nvolume is now -> 43\nvolume is
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nvolume is now -> 47\nvolume is now -> 48\nvolume is now
-> 49\nvolume is now -> 50\nvolume is now -> 51\nvolume is
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nvolume is now -> 55\nvolume is now -> 56\nvolume is now
-> 57\nvolume is now -> 58\nvolume is now -> 59\nvolume is
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nvolume is now -> 63\nvolume is now -> 64\nvolume is now
-> 65\nvolume is now -> 66\nvolume is now -> 67\nvolume is
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nvolume is now -> 71\nvolume is now -> 72\nvolume is now
-> 73\nvolume is now -> 74\nvolume is now -> 75\nvolume is
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nvolume is now -> 79\nvolume is now -> 80\nvolume is now
-> 81\nvolume is now -> 82\nvolume is now -> 83\nvolume is
 now -> 84\nvolume is now -> 85\nvolume is now -> 86\
nvolume is now -> 87\nvolume is now -> 88\nvolume is now
-> 89\nvolume is now -> 90\nvolume is now -> 91\nvolume is
 now -> 92\nvolume is now -> 93\nvolume is now -> 94\
nvolume is now -> 95\nvolume is now -> 96\nvolume is now
-> 97\nvolume is now -> 98\nvolume is now -> 99\nvolume is
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nvolume is now -> 100\nvolume is now -> 100\nvolume is now
 -> 100\nvolume is now -> 100\nvolume is now -> 100\
nvolume is now -> 100\nvolume is now -> 100\nvolume is now
 -> 100\nDecrease volume 110 times.\nvolume is now -> 99\
nvolume is now -> 98\nvolume is now -> 97\nvolume is now
-> 96\nvolume is now -> 95\nvolume is now -> 94\nvolume is
 now -> 93\nvolume is now -> 92\nvolume is now -> 91\
nvolume is now -> 90\nvolume is now -> 89\nvolume is now
-> 88\nvolume is now -> 87\nvolume is now -> 86\nvolume is
 now -> 85\nvolume is now -> 84\nvolume is now -> 83\
nvolume is now -> 82\nvolume is now -> 81\nvolume is now
-> 80\nvolume is now -> 79\nvolume is now -> 78\nvolume is
 now -> 77\nvolume is now -> 76\nvolume is now -> 75\
nvolume is now -> 74\nvolume is now -> 73\nvolume is now
-> 72\nvolume is now -> 71\nvolume is now -> 70\nvolume is
 now -> 69\nvolume is now -> 68\nvolume is now -> 67\
nvolume is now -> 66\nvolume is now -> 65\nvolume is now
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            -> 64\nvolume is now -> 63\nvolume is now -> 62\nvolume is
             now -> 61\nvolume is now -> 60\nvolume is now -> 59\
            nvolume is now -> 58\nvolume is now -> 57\nvolume is now
            -> 56\nvolume is now -> 55\nvolume is now -> 54\nvolume is
             now -> 53\nvolume is now -> 52\nvolume is now -> 51\
            nvolume is now -> 50\nvolume is now -> 49\nvolume is now
            -> 48\nvolume is now -> 47\nvolume is now -> 46\nvolume is
             now -> 45\nvolume is now -> 44\nvolume is now -> 43\
            nvolume is now -> 42\nvolume is now -> 41\nvolume is now
            -> 40\nvolume is now -> 39\nvolume is now -> 38\nvolume is
             now -> 37\nvolume is now -> 36\nvolume is now -> 35\
            nvolume is now -> 34\nvolume is now -> 33\nvolume is now
            -> 32\nvolume is now -> 31\nvolume is now -> 30\nvolume is
             now -> 29\nvolume is now -> 28\nvolume is now -> 27\
            nvolume is now -> 26\nvolume is now -> 25\nvolume is now
            -> 24\nvolume is now -> 23\nvolume is now -> 22\nvolume is
             now -> 21\nvolume is now -> 20\nvolume is now -> 19\
            nvolume is now -> 18\nvolume is now -> 17\nvolume is now
            -> 16\nvolume is now -> 15\nvolume is now -> 14\nvolume is
             now -> 13\nvolume is now -> 12\nvolume is now -> 11\
            nvolume is now -> 10\nvolume is now -> 9\nvolume is now ->
             8\nvolume is now -> 7\nvolume is now -> 6\nvolume is now
            -> 5\nvolume is now -> 4\nvolume is now -> 3\nvolume is
            now -> 2\nvolume is now -> 1\nvolume is now -> 0\nvolume
            is now -> 0\nvolume is now -> 0\nvolume is now -> 0\
            nvolume is now -> 0\nvolume is now -> 0\nvolume is now ->
            0\nvolume is now -> 0\nvolume is now -> 0\nvolume is now
            -> 0\nvolume is now -> 0\n\n### Test Next/Prev ###\n- Wrap
             around to last:\nplaying song number 4: Media 2 (4 sec)\n
            - Wrap around to first:\nplaying song number 1: Media 1 (1
             sec)\n- Switch through all entries:\nplaying song number
            2: Media 2 (2 sec)\nplaying song number 3: Media 3 (3 sec)
            \nplaying song number 4: Media 2 (4 sec)\nplaying song
            number 1: Media 1 (1 sec)\nplaying song number 2: Media 2
            (2 sec)\n\n### Test Select ###\n-- Search for \"Does not
            exist\":\nsong: Does not exist not found!\nplaying song
            number 2: Media 2 (2 sec)\nstop song: Media 2 (2 sec)\n\n
            -- Search for \"Media 1\":\nplaying song number 1: Media 1
             (1 sec)\nstop song: Media 1 (1 sec)\n\n-- Search for \"
            Media 2\":\nplaying song number 2: Media 2 (2 sec)\nstop
            song: Media 2 (2 sec)\n\n-- Search for \"Media 2\" -
            Should find the same one again:\nplaying song number 2:
            Media 2 (2 sec)\nstop song: Media 2 (2 sec)\n");
}

////////////////////////////////////////////////

TEST(PlayerClient, nullptr) {
    try {
        PlayerClient const client = PlayerClient{ nullptr };
        EXPECT_EQ(1, 0); // Fail.
    }
    catch (std::invalid_argument const& ex) {
        EXPECT_STREQ(ex.what(), "Provided player is nullptr");
    }
```

```
572
573     PlayerClient client = PlayerClient{ std::make_unique<
            MusicPlayerAdapter>(std::move(std::make_unique<MusicPlayer
            >())) };
574     try {
575         client.SetPlayer(nullptr);
576         EXPECT_EQ(1, 0); // Fail.
577     }
578     catch (std::invalid_argument const& ex) {
579         EXPECT_STREQ(ex.what(), "Provided player is nullptr");
580     }
581 }
582
583 TEST(PlayerClient, SetPlayer) {
584     std::string output = "";
585
586     PlayerClient client = PlayerClient{ std::make_unique<
            MusicPlayerAdapter>(std::move(std::make_unique<MusicPlayer
            >())) };
587
588     CaptureStdout();
589     client.Run();
590     output = GetCapturedStdout();
591     cout << output;
592     EXPECT_EQ(output, "### Test Stop/Play ###\nno song in playlist
            !\nno song in playlist!\nIncrease volume 110 times.\
            nvolume is now -> 16\nvolume is now -> 17\nvolume is now
            -> 18\nvolume is now -> 19\nvolume is now -> 20\nvolume is
             now -> 21\nvolume is now -> 22\nvolume is now -> 23\
            nvolume is now -> 24\nvolume is now -> 25\nvolume is now
            -> 26\nvolume is now -> 27\nvolume is now -> 28\nvolume is
             now -> 29\nvolume is now -> 30\nvolume is now -> 31\
            nvolume is now -> 32\nvolume is now -> 33\nvolume is now
            -> 34\nvolume is now -> 35\nvolume is now -> 36\nvolume is
             now -> 37\nvolume is now -> 38\nvolume is now -> 39\
            nvolume is now -> 40\nvolume is now -> 41\nvolume is now
            -> 42\nvolume is now -> 43\nvolume is now -> 44\nvolume is
             now -> 45\nvolume is now -> 46\nvolume is now -> 47\
            nvolume is now -> 48\nvolume is now -> 49\nvolume is now
            -> 50\nvolume is now -> 51\nvolume is now -> 52\nvolume is
             now -> 53\nvolume is now -> 54\nvolume is now -> 55\
            nvolume is now -> 56\nvolume is now -> 57\nvolume is now
            -> 58\nvolume is now -> 59\nvolume is now -> 60\nvolume is
             now -> 61\nvolume is now -> 62\nvolume is now -> 63\
            nvolume is now -> 64\nvolume is now -> 65\nvolume is now
            -> 66\nvolume is now -> 67\nvolume is now -> 68\nvolume is
             now -> 69\nvolume is now -> 70\nvolume is now -> 71\
            nvolume is now -> 72\nvolume is now -> 73\nvolume is now
            -> 74\nvolume is now -> 75\nvolume is now -> 76\nvolume is
             now -> 77\nvolume is now -> 78\nvolume is now -> 79\
            nvolume is now -> 80\nvolume is now -> 81\nvolume is now
            -> 82\nvolume is now -> 83\nvolume is now -> 84\nvolume is
             now -> 85\nvolume is now -> 86\nvolume is now -> 87\
            nvolume is now -> 88\nvolume is now -> 89\nvolume is now
            -> 90\nvolume is now -> 91\nvolume is now -> 92\nvolume is
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 now -> 93\nvolume is now -> 94\nvolume is now -> 95\
nvolume is now -> 96\nvolume is now -> 97\nvolume is now
-> 98\nvolume is now -> 99\nvolume is now -> 100\nvolume
is now -> 100\nvolume is now -> 100\nvolume is now -> 100\
nvolume is now -> 100\nvolume is now -> 100\nvolume is now
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nvolume is now -> 100\nvolume is now -> 100\nvolume is now
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nvolume is now -> 100\nvolume is now -> 100\nvolume is now
 -> 100\nvolume is now -> 100\nvolume is now -> 100\
nvolume is now -> 100\nDecrease volume 110 times.\nvolume
is now -> 99\nvolume is now -> 98\nvolume is now -> 97\
nvolume is now -> 96\nvolume is now -> 95\nvolume is now
-> 94\nvolume is now -> 93\nvolume is now -> 92\nvolume
is now -> 91\nvolume is now -> 90\nvolume is now -> 89\
nvolume is now -> 88\nvolume is now -> 87\nvolume is now
-> 86\nvolume is now -> 85\nvolume is now -> 84\nvolume
is now -> 83\nvolume is now -> 82\nvolume is now -> 81\
nvolume is now -> 80\nvolume is now -> 79\nvolume is now
-> 78\nvolume is now -> 77\nvolume is now -> 76\nvolume
is now -> 75\nvolume is now -> 74\nvolume is now -> 73\
nvolume is now -> 72\nvolume is now -> 71\nvolume is now
-> 70\nvolume is now -> 69\nvolume is now -> 68\nvolume
is now -> 67\nvolume is now -> 66\nvolume is now -> 65\
nvolume is now -> 64\nvolume is now -> 63\nvolume is now
-> 62\nvolume is now -> 61\nvolume is now -> 60\nvolume
is now -> 59\nvolume is now -> 58\nvolume is now -> 57\
nvolume is now -> 56\nvolume is now -> 55\nvolume is now
-> 54\nvolume is now -> 53\nvolume is now -> 52\nvolume
is now -> 51\nvolume is now -> 50\nvolume is now -> 49\
nvolume is now -> 48\nvolume is now -> 47\nvolume is now
-> 46\nvolume is now -> 45\nvolume is now -> 44\nvolume
is now -> 43\nvolume is now -> 42\nvolume is now -> 41\
nvolume is now -> 40\nvolume is now -> 39\nvolume is now
-> 38\nvolume is now -> 37\nvolume is now -> 36\nvolume
is now -> 35\nvolume is now -> 34\nvolume is now -> 33\
nvolume is now -> 32\nvolume is now -> 31\nvolume is now
-> 30\nvolume is now -> 29\nvolume is now -> 28\nvolume
is now -> 27\nvolume is now -> 26\nvolume is now -> 25\
nvolume is now -> 24\nvolume is now -> 23\nvolume is now
-> 22\nvolume is now -> 21\nvolume is now -> 20\nvolume
is now -> 19\nvolume is now -> 18\nvolume is now -> 17\
nvolume is now -> 16\nvolume is now -> 15\nvolume is now
-> 14\nvolume is now -> 13\nvolume is now -> 12\nvolume
is now -> 11\nvolume is now -> 10\nvolume is now -> 9\
nvolume is now -> 8\nvolume is now -> 7\nvolume is now ->
6\nvolume is now -> 5\nvolume is now -> 4\nvolume is now
-> 3\nvolume is now -> 2\nvolume is now -> 1\nvolume is
now -> 0\nvolume is now -> 0\nvolume is now -> 0\nvolume
is now -> 0\nvolume is now -> 0\nvolume is now -> 0\
nvolume is now -> 0\nvolume is now -> 0\nvolume is now ->
0\nvolume is now -> 0\nvolume is now -> 0\nvolume is now
-> 0\nvolume is now -> 0\nvolume is now -> 0\nvolume is
now -> 0\nvolume is now -> 0\nvolume is now -> 0\nvolume
is now -> 0\n\n### Test Next/Prev ###\n- Wrap around to
```

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        last:\nno song in playlist!\n- Wrap around to first:\nno
        song in playlist!\n- Switch through all entries:\nno song
        in playlist!\nno song in playlist!\nno song in playlist!\
        nno song in playlist!\nno song in playlist!\n\n### Test
        Select ###\n-- Search for \"Does not exist\":\nsong: Does
        not exist not found!\nno song in playlist!\nno song in
        playlist!\n\n-- Search for \"Media 1\":\nsong: Media 1 not
         found!\nno song in playlist!\nno song in playlist!\n\n--
        Search for \"Media 2\":\nsong: Media 2 not found!\nno song
         in playlist!\nno song in playlist!\n\n-- Search for \"
        Media 2\" - Should find the same one again:\nsong: Media 2
         not found!\nno song in playlist!\nno song in playlist!\n"
        );
593
594     client.SetPlayer(std::make_unique<VideoPlayerAdapter>(std::
            move(std::make_unique<VideoPlayer>())));
595
596     CaptureStdout();
597     client.Run();
598     output = GetCapturedStdout();
599     cout << output;
600     EXPECT_EQ(output, "### Test Stop/Play ###\nno video in
            playlist!\nno video in playlist!\nIncrease volume 110
            times.\nvolume is now -> 9\nvolume is now -> 10\nvolume is
             now -> 11\nvolume is now -> 12\nvolume is now -> 13\
            nvolume is now -> 14\nvolume is now -> 15\nvolume is now
            -> 16\nvolume is now -> 17\nvolume is now -> 18\nvolume is
             now -> 19\nvolume is now -> 20\nvolume is now -> 21\
            nvolume is now -> 22\nvolume is now -> 23\nvolume is now
            -> 24\nvolume is now -> 25\nvolume is now -> 26\nvolume is
             now -> 27\nvolume is now -> 28\nvolume is now -> 29\
            nvolume is now -> 30\nvolume is now -> 31\nvolume is now
            -> 32\nvolume is now -> 33\nvolume is now -> 34\nvolume is
             now -> 35\nvolume is now -> 36\nvolume is now -> 37\
            nvolume is now -> 38\nvolume is now -> 39\nvolume is now
            -> 40\nvolume is now -> 41\nvolume is now -> 42\nvolume is
             now -> 43\nvolume is now -> 44\nvolume is now -> 45\
            nvolume is now -> 46\nvolume is now -> 47\nvolume is now
            -> 48\nvolume is now -> 49\nvolume is now -> 50\nvolume is
             now -> 50\nvolume is now -> 50\nvolume is now -> 50\
            nvolume is now -> 50\nvolume is now -> 50\nvolume is now
            -> 50\nvolume is now -> 50\nvolume is now -> 50\nvolume is
             now -> 50\nvolume is now -> 50\nvolume is now -> 50\
            nvolume is now -> 50\nvolume is now -> 50\nvolume is now
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            nvolume is now -> 50\nvolume is now -> 50\nvolume is now
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             now -> 50\nvolume is now -> 50\nvolume is now -> 50\
            nvolume is now -> 50\nvolume is now -> 50\nvolume is now
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             now -> 50\nvolume is now -> 50\nvolume is now -> 50\
            nvolume is now -> 50\nvolume is now -> 50\nvolume is now
            -> 50\nvolume is now -> 50\nvolume is now -> 50\nvolume is
             now -> 50\nvolume is now -> 50\nvolume is now -> 50\
            nvolume is now -> 50\nvolume is now -> 50\nvolume is now
            -> 50\nvolume is now -> 50\nvolume is now -> 50\nvolume is
             now -> 50\nvolume is now -> 50\nvolume is now -> 50\
```

```
        nvolume is now -> 50\nvolume is now -> 50\nvolume is now
        -> 50\nvolume is now -> 50\nvolume is now -> 50\nvolume is
         now -> 50\nvolume is now -> 50\nvolume is now -> 50\
        nvolume is now -> 50\nvolume is now -> 50\nvolume is now
        -> 50\nvolume is now -> 50\nvolume is now -> 50\nvolume is
         now -> 50\nvolume is now -> 50\nvolume is now -> 50\
        nvolume is now -> 50\nvolume is now -> 50\nvolume is now
        -> 50\nvolume is now -> 50\nvolume is now -> 50\nvolume is
         now -> 50\nvolume is now -> 50\nvolume is now -> 50\
        nvolume is now -> 50\nDecrease volume 110 times.\nvolume
        is now -> 49\nvolume is now -> 48\nvolume is now -> 47\
        nvolume is now -> 46\nvolume is now -> 45\nvolume is now
        -> 44\nvolume is now -> 43\nvolume is now -> 42\nvolume is
         now -> 41\nvolume is now -> 40\nvolume is now -> 39\
        nvolume is now -> 38\nvolume is now -> 37\nvolume is now
        -> 36\nvolume is now -> 35\nvolume is now -> 34\nvolume is
         now -> 33\nvolume is now -> 32\nvolume is now -> 31\
        nvolume is now -> 30\nvolume is now -> 29\nvolume is now
        -> 28\nvolume is now -> 27\nvolume is now -> 26\nvolume is
         now -> 25\nvolume is now -> 24\nvolume is now -> 23\
        nvolume is now -> 22\nvolume is now -> 21\nvolume is now
        -> 20\nvolume is now -> 19\nvolume is now -> 18\nvolume is
         now -> 17\nvolume is now -> 16\nvolume is now -> 15\
        nvolume is now -> 14\nvolume is now -> 13\nvolume is now
        -> 12\nvolume is now -> 11\nvolume is now -> 10\nvolume is
         now -> 9\nvolume is now -> 8\nvolume is now -> 7\nvolume
        is now -> 6\nvolume is now -> 5\nvolume is now -> 4\
        nvolume is now -> 3\nvolume is now -> 2\nvolume is now ->
        1\nvolume is now -> 0\n\n### Test Next/Prev ###\n- Wrap
        around to last:\nno video in playlist!\n- Wrap around to
        first:\nno video in playlist!\n- Switch through all
        entries:\nno video in playlist!\nno video in playlist!\nno
         video in playlist!\nno video in playlist!\nno video in
        playlist!\n\n### Test Select ###\n-- Search for \"Does not
         exist\":\nvideo: Does not exist not found!\nno video in
        playlist!\nno video in playlist!\n\n-- Search for \"Media
        1\":\nvideo: Media 1 not found!\nno video in playlist!\nno
         video in playlist!\n\n-- Search for \"Media 2\":\nvideo:
        Media 2 not found!\nno video in playlist!\nno video in
        playlist!\n\n-- Search for \"Media 2\" - Should find the
        same one again:\nvideo: Media 2 not found!\nno video in
        playlist!\nno video in playlist!\n");
601 }
```